

BEC304- NETWORK ANALYSIS

MODULE-2 NETWORK THEOREMS

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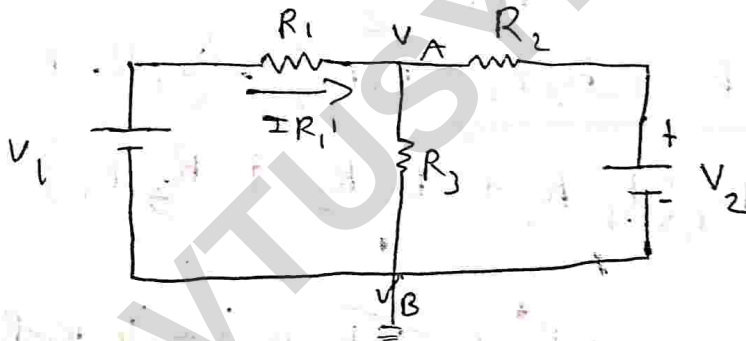
Superposition Theorem :

statement :

In any linear Bilateral network containing more than one independent sources (V or I). The response (current through or voltage across) in any element is the algebraic sum of all the responses produced by the sources acting alone.

Note : To find individual responses voltage sources are shorted and current sources are open ~~at~~ circuited :

• consider the following network



let the response is current through R_1 (I_{R1})

$$\frac{V_A - V_1}{R_1} + \frac{V_A}{R_3} + \frac{V_A - V_2}{R_2} = 0$$

$$V_A \left(\frac{1}{R_1} + \frac{1}{R_3} + \frac{1}{R_2} \right) = \frac{V_1}{R_1} + \frac{V_2}{R_2}$$

$$V_A \left(\frac{1}{R_1} + \frac{1}{R_3} + \frac{1}{R_2} \right) = \frac{V_1 R_2 + V_2 R_1}{R_1 + R_2}$$

$$V_A \left(\frac{R_3 R_2 + R_1 R_2 + R_1 R_3}{R_1 R_2 R_3} \right) = \left(\frac{V_1 R_2 + V_2 R_1}{R_1 R_2} \right)$$

$$V_A = \left(\frac{\cancel{R_1} \cancel{R_2} R_3}{R_3 R_2 + R_1 R_2 + R_1 R_3} \right) \left(\frac{V_1 R_2 + V_2 R_1}{\cancel{R_1} \cancel{R_2}} \right)$$

$$V_A = \frac{V_1 R_2 R_3 + V_2 R_1 R_3}{R_1 R_2 + R_3 R_2 + R_1 R_3} \quad \dots \quad (1)$$

The response is $I_{R_1} = \frac{V_1 - V_A}{R_1} \quad \dots \quad (2)$

sub. (1) in (2)

$$I_{R_1} = \frac{V_1 - \frac{V_1 R_2 R_3 + V_2 R_1 R_3}{R_1 R_2 + R_3 R_2 + R_1 R_3}}{R_1}$$

$$I_{R_1} = \frac{V_1 (R_1 R_2 + R_3 R_2 + R_1 R_3) - (V_1 R_2 R_3 + V_2 R_1 R_3)}{R_1 (R_1 R_2 + R_2 R_3 + R_3 R_1)}$$

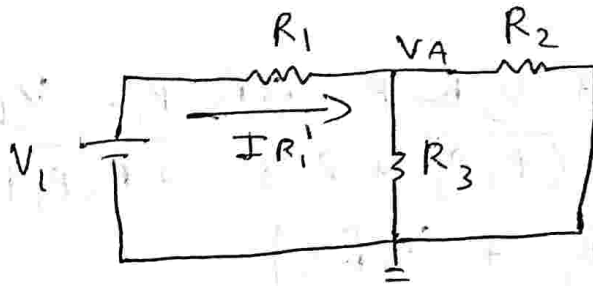
$$I_{R_1} = \frac{(V_1 \cancel{R_1} R_2 + \cancel{V_1} R_3 R_2 + \cancel{V_1} R_1 R_3) - (V_1 \cancel{R_2} R_3 + V_2 R_1 R_3)}{R_1 (R_1 R_2 + R_2 R_3 + (R_1 R_3))}$$

$$I_{R_1} = \frac{\cancel{R_1} (V_1 R_2 + V_1 R_3 - V_2 R_3)}{\cancel{R_1} (R_1 R_2 + R_2 R_3 + R_3 R_1)}$$

$$I_{R_1} = \frac{V_1 (R_2 + R_3 - V_2 R_3)}{R_1 R_2 + R_2 R_3 + R_3 R_1} \quad \dots \quad (A)$$

To prove - $IR_1 = IR_1' + IR_1''$

con 1: V_1 acting alone



apply KCL

$$\frac{V_A - V_1}{R_1} + \frac{V_A}{R_3} + \frac{V_A}{R_2} = 0$$

$$V_A \left(\frac{1}{R_1} + \frac{1}{R_3} + \frac{1}{R_2} \right) = \frac{V_1}{R_1}$$

$$V_A \left(\frac{R_1 R_2 + R_2 R_3 + R_3 R_1}{R_1 R_2 R_3} \right) = \frac{V_1}{R_1}$$

$$V_A = \frac{V_1 \cancel{R_1} R_2 R_3}{\cancel{R_1} (R_1 R_2 + R_2 R_3 + R_3 R_1)}$$

$$\therefore V_A = \frac{V_1 R_2 R_3}{(R_1 R_2 + R_2 R_3 + R_3 R_1)} \quad \text{--- (3)}$$

from the ckt

$$IR_1' = \frac{V_1 - V_A}{R_1}$$

$$IR_1' = V_1 - \frac{V_1 R_2 R_3}{R_1 R_2 + R_2 R_3 + R_3 R_1}$$

$$IR_1' = \frac{V_1(R_1R_2 + R_2R_3 + R_3R_1) - V_1(R_2R_3)}{R_1(R_1R_2 + R_2R_3 + R_3R_1)}$$

$$IR_1' = \frac{V_1R_1R_2 + \cancel{V_1R_2R_3} + \cancel{V_1R_3R_1} - \cancel{V_1R_2R_3}}{R_1(R_1R_2 + R_2R_3 + R_3R_1)}$$

$$IR_1' = \frac{\cancel{R_1}(V_1R_2 + V_1R_3)}{\cancel{R_1}(R_1R_2 + R_2R_3 + R_3R_1)}$$

$$IR_1' = \frac{(V_1(R_2 + R_3))}{R_1R_2 + R_2R_3 + R_3R_1} \dots \textcircled{5}$$

case 2 :



apply KCL

$$\frac{V_A}{R_1} + \frac{V_A}{R_3} + \frac{V_A - V_2}{R_2}$$

$$V_A \left(\frac{R_1R_2 + R_2R_3 + R_3R_1}{R_1R_2R_3} \right) = \frac{V_2}{R_2}$$

$$V_A = \frac{V_2}{\cancel{R_2}} \left(\frac{\cancel{R_1R_2R_3}}{R_1R_2 + R_2R_3 + R_3R_1} \right)$$

$$V_A = \frac{V_2 R_1 R_3}{R_1R_2 + R_2R_3 + R_3R_1} \dots \textcircled{6}$$

The voltage $I R_1' = 0 - V_A$

$$I R_1' = 0 - \frac{V_2 R_1 R_3}{R_1 R_2 + R_2 R_3 + R_3 R_1}$$

$$I R_1'' = \frac{-V_2 R_1 R_3}{R_1 (R_1 R_2 + R_2 R_3 + R_3 R_1)}$$

$$I R_1'' = \frac{-V_2 R_3}{R_1 R_2 + R_2 R_3 + R_3 R_1} \quad \text{--- (7)}$$

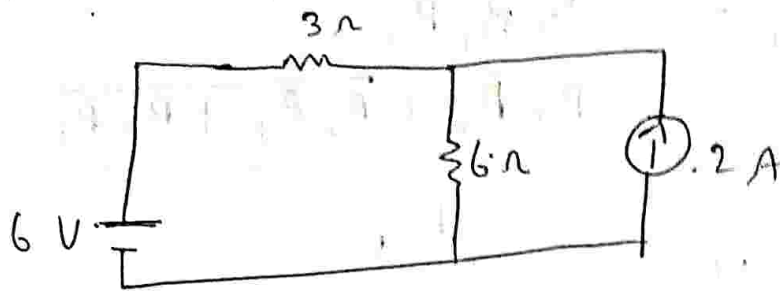
$$I R_1 = I R_1' + I R_1''$$

$$I R_1 = \frac{V_1 (R_2 + R_3)}{R_1 R_2 + R_2 R_3 + R_3 R_1} - \frac{V_2 R_3}{R_1 R_2 + R_2 R_3 + R_3 R_1}$$

$$\therefore I R_1 = \frac{V_1 (R_2 + R_3) - V_2 R_3}{R_1 R_2 + R_2 R_3 + R_3 R_1} \quad \text{--- (8)}$$

$$\therefore A = B_{11}$$

1) Find the current through 6Ω resistance using super position theorem



current through 6Ω $I_{6\Omega} = ?$
according to super position

$$I_{6\Omega} = I_{6\Omega}^I + I_{6\Omega}^{II}$$

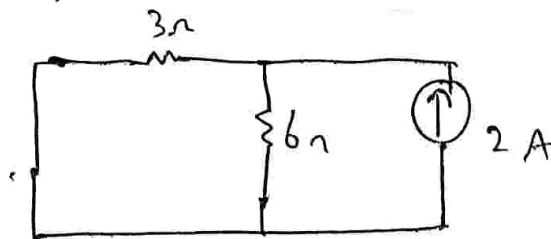
Case 1:
 $6V$ acting alone



$$I_{6\Omega}^I = \frac{6}{9} = 0.66A$$

Case 2:

$2A$ current source acting alone
short ckt $6V$ (V) source.



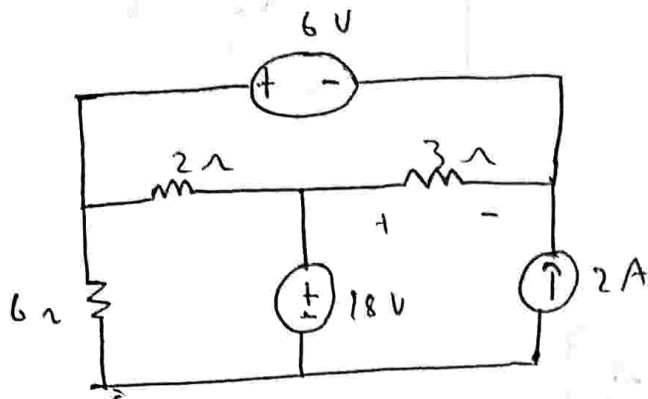
apply IPRule.

$$I_{6\Omega}^{II} = \frac{2(3)}{6+3} = \frac{6}{9} = 0.66A$$

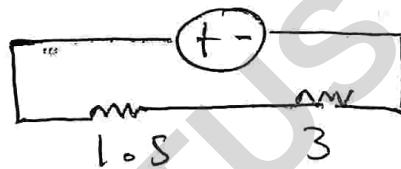
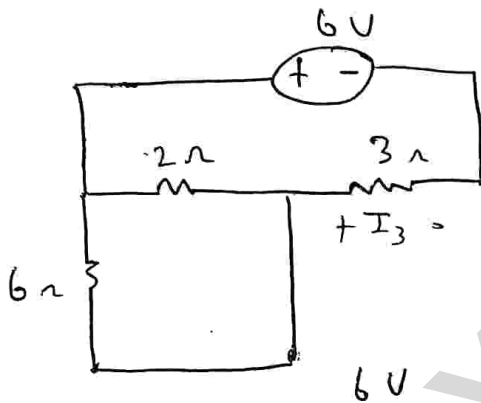
$$I_{6\Omega} = 0.66A + 0.66A$$

$$I_{6\Omega} = 1.32A$$

2) Find the voltage across 3Ω resistor using superposition theorem



Soln: case ①
6V acting alone. { 18V is shorted 2A open



$$I_3 = \frac{6}{1.5 + 3}$$

$$I_3 = 1.33A$$

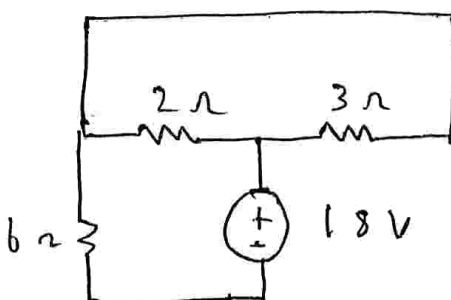
$$V^1 = I R$$

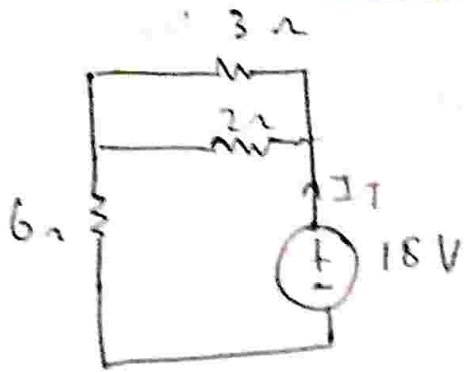
$$V^1 = 1.33 \times 3$$

$$V^1 = 4V$$

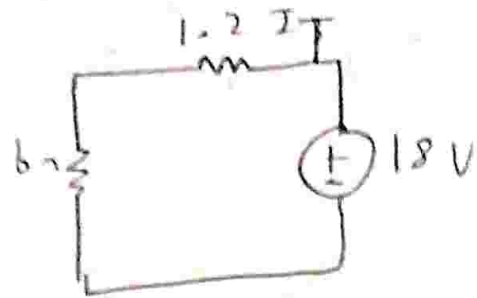
case ②

consider 18V alone short 6V { open 2A





$\Rightarrow$



$$I_T = \frac{18}{7.2}$$

$$I_T = 2.5 A$$

$$I_3 = \frac{I_T \times 2}{2 + 3}$$

$$I_3 = \frac{2.5 \times 2}{5}$$

$$I_3 = 1 A$$

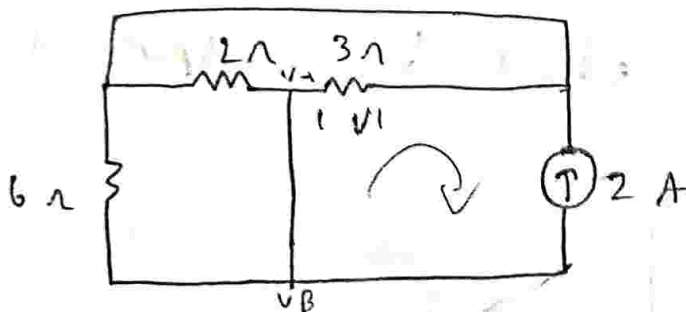
$$V = IR$$

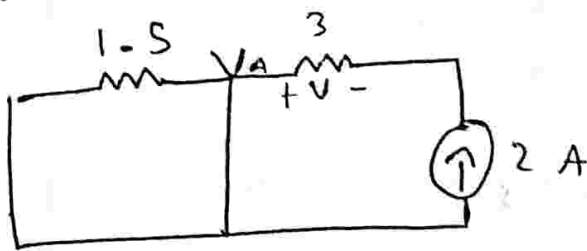
$$V = 1 \times 3$$

$$V'' = 3 V$$

Case ③

2A acting alone short 6 ohm & 18V





applying KCL

$$\frac{V_A - 0}{1.5} + \frac{V_A - 0}{3} = -2A$$

$$V_A \left(\frac{1}{1.5} + \frac{1}{3} \right) = -2A$$

$$\therefore V_A (1) = -2A$$

$$\boxed{V_A = -2V}$$

$$\text{or } V_{III} = -2V$$

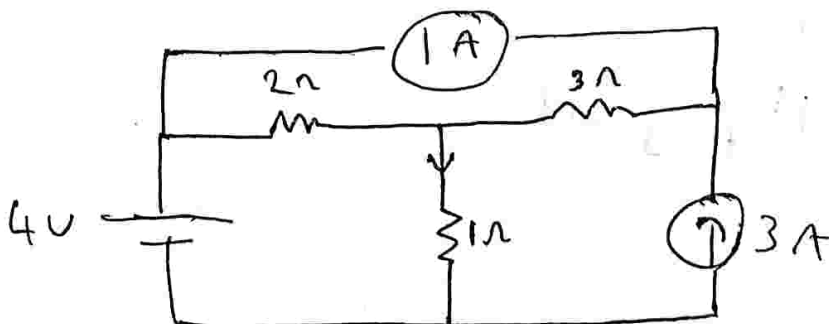
$$\therefore V_{3\Omega} = V_I + V_{II} + V_{III}$$

$$V_{3\Omega} = 4 + 3 - 2$$

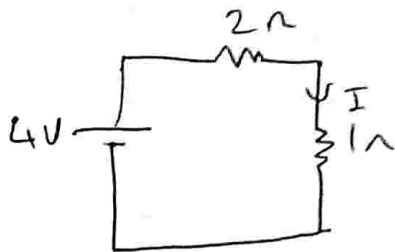
$$\boxed{V_{3\Omega} = 5V}$$

//

3) Find the current through 1Ω resistor using superposition theorem.



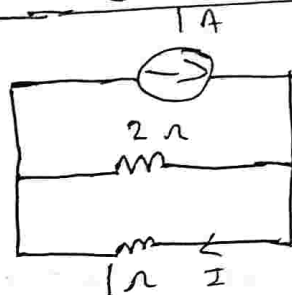
consider 4V alone



$$I' = \frac{V}{R} = \frac{4}{2+1} = 1.33 \text{ A}$$

$$I' = 1.33 \text{ A}$$

consider 1A current source: short (3A current)

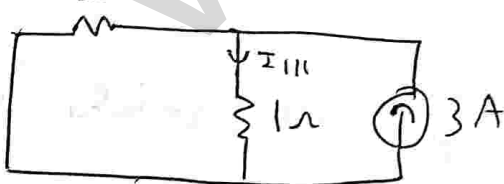


apply current divider

$$I'' = \frac{1 \times (2)}{2+1} = 0.666$$

$$I'' = 0.666 \text{ A}$$

consider 3A current

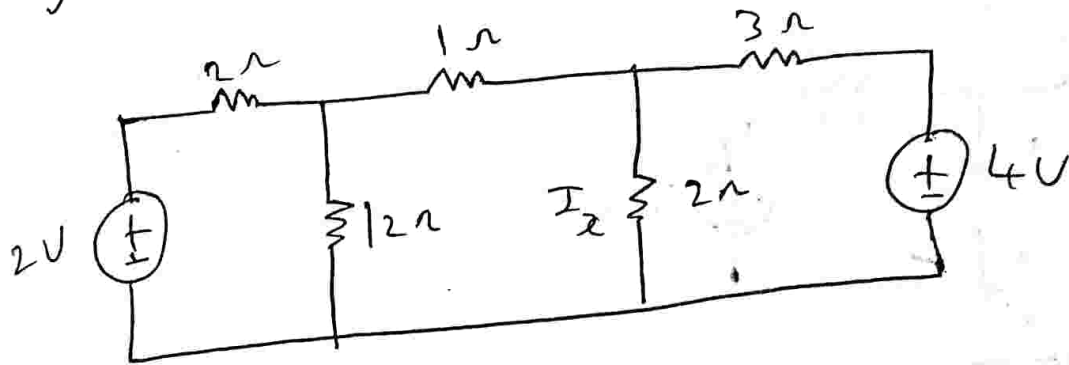


$$I''' = \frac{3(2)}{2+1} = 2 \text{ A}$$

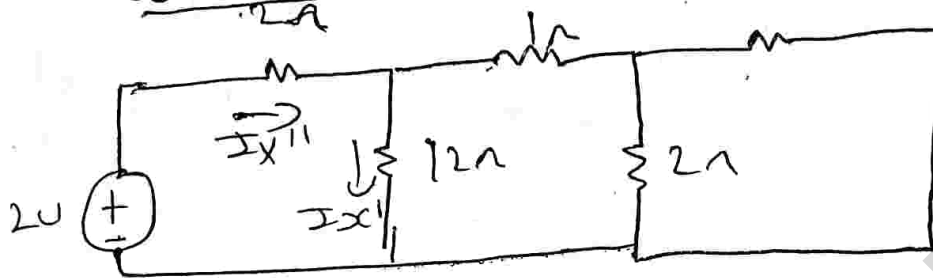
$$I = I' + I'' + I'''$$

$$I = 4 \text{ A}$$

4) Find the current in $R = 2\Omega$



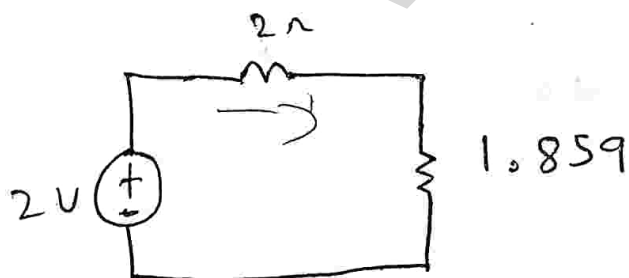
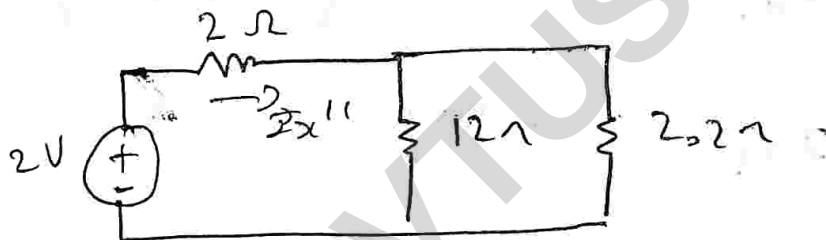
consider 2V ~~source~~ alone:



$$3 \parallel 2 + 1\Omega$$

$$0.2 + 1$$

$$2.2\Omega$$



$$I_x'' = \frac{2}{2 + 1.859 + 2} = 0.5182A$$

$$I_{x'} = 0.854 \times I_x''$$

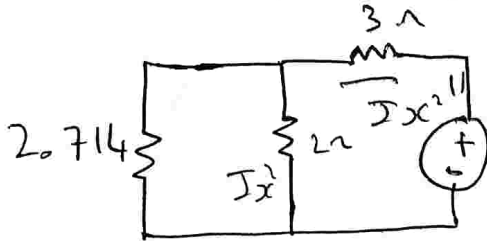
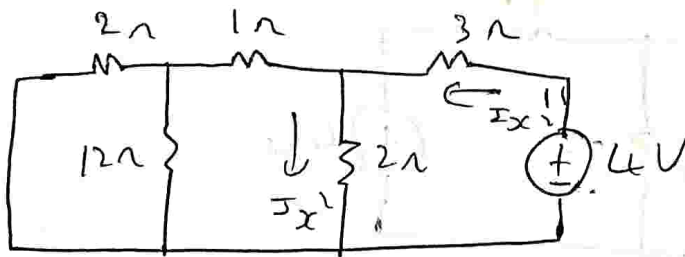
$$I_{x'} = 0.854 \times 0.5182$$

$$I_{x'} = \frac{12 \times I_x''}{12 + 2.2}$$

$$I_{x'} = 0.45 I_x''$$

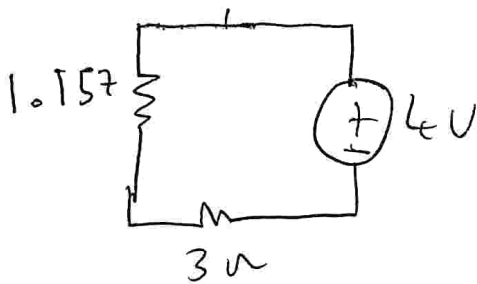
$$\therefore I_x = 0.2627A$$

Case (ii) let $4V$ ~~be~~ be acting alone



$$I_{x2} = \frac{2.714 I_{x1}}{2 + 2.714}$$

$$I_{x2} = 0.572 I_{x1}$$



$$I_{x1} = \frac{V}{R} = \frac{4}{3 + 1.157}$$

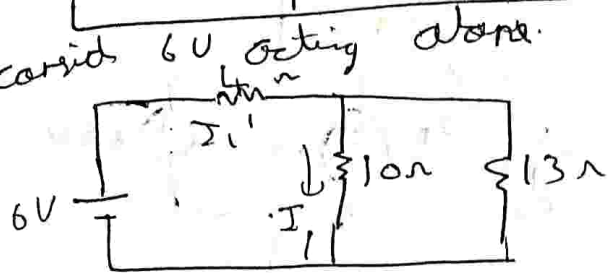
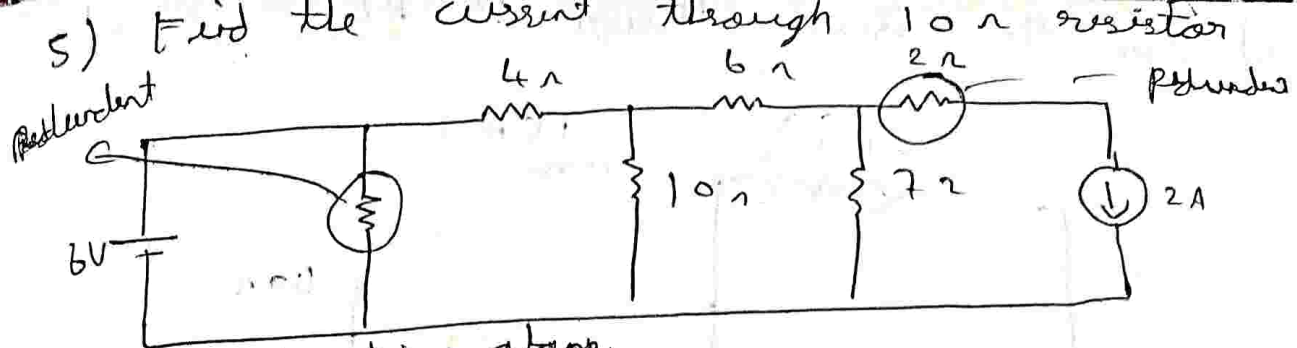
$$I_{x1} = 0.572 \left(\frac{4}{3 + 1.157} \right)$$

$$I_{x1} = 0.05540$$

$$I_{xc} = I_{x1} + I_{x2}$$

$$I_{xc} = 0.2627 + 0.5540$$

$$I_{xc} = 0.8167 A$$



$$I_1 = \frac{13 \times I_1'}{13 + 10}$$

$$I_1 = 0.5652 I_1'$$

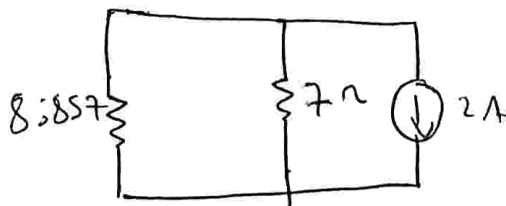
$$I_1' = \frac{6}{4 + 5.652}$$

$$I_1' = 0.6216$$

$$I_1 = 0.5652 (0.6216)$$

$$I_1 = 0.35A$$

Case 2 consider 2A acting alone.



$$I_2 = -\frac{I_2' \times 4}{4 + 10}$$

$$I_2 = -0.485 I_2'$$

$$I_2 = -0.2857 \times 0.882$$

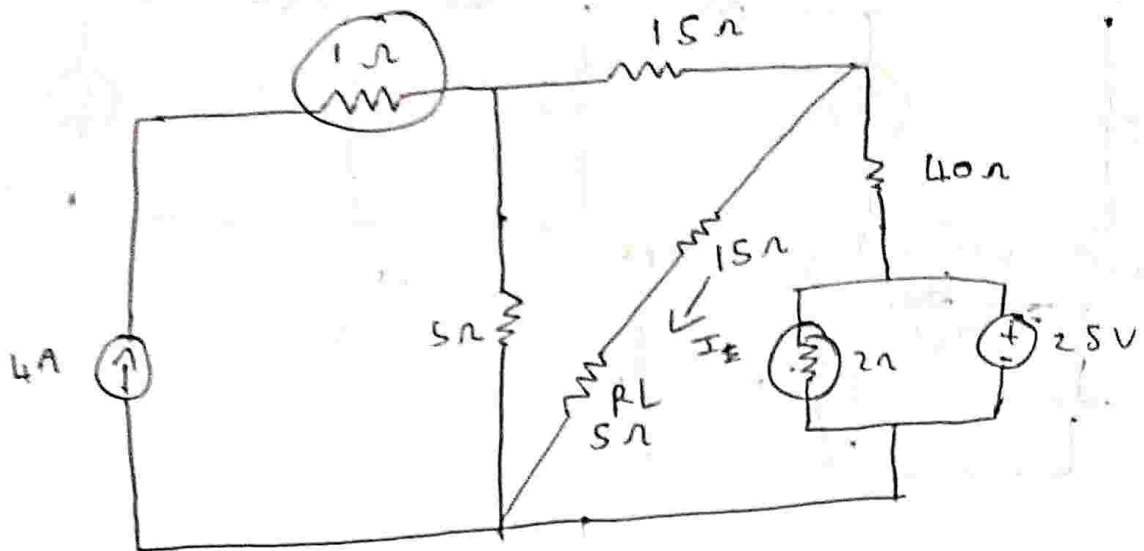
$$I_2 = -0.2857 \times 0.882$$

$$I = 0.099A$$

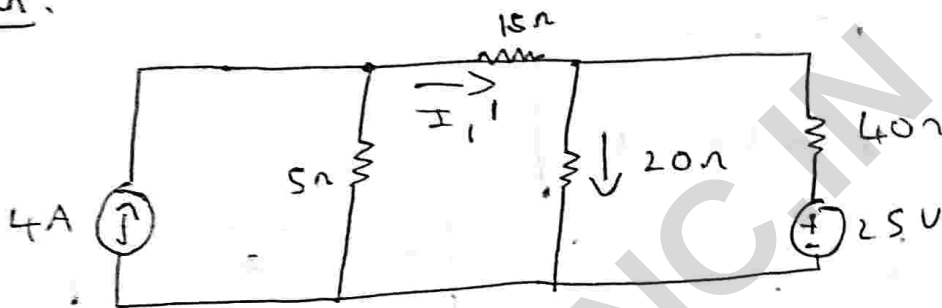
$$I_2' = \frac{7 \times 2}{8.857 + 7}$$

$$I_2' = 0.882A$$

b) Find the current through ~~the~~ RL



Soln:

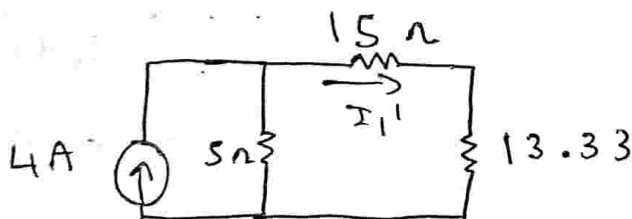


4A Acting alone



$$I_1 = \frac{I_{11} \times 40}{40 + 20}$$

$$I_1 = 0.66 I_{11}$$



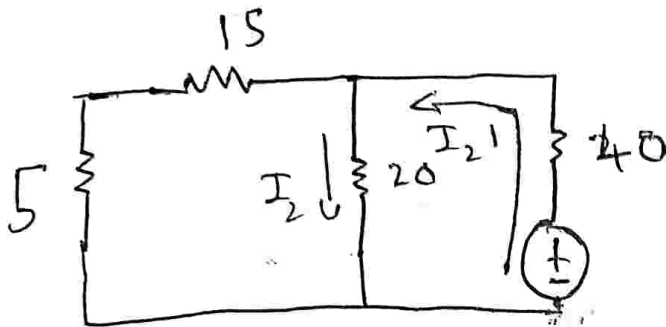
$$I_{11} = \frac{4 \times 5}{5 + (15 \parallel 13.33)}$$

$$I_1' = 0.64$$

$$I_1 = 0.6 \times 0.66$$

$$I_1 = 0.4 \text{ A}$$

25V acting alone.



$$I_2 = \frac{I_2' \cdot 20}{20 + 20}$$

$$I_2 = 0.5 I_2'$$

$$I_2' = \frac{V}{R} = \frac{25}{40 + \left(\frac{20 \times 20}{40} \right)}$$

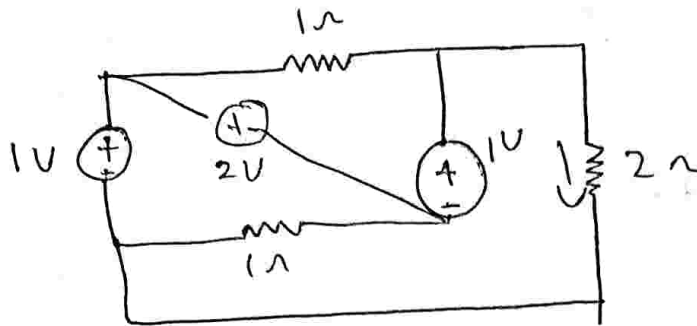
$$I_2' = 0.5 \text{ A}$$

$$I_2 = 0.25 \text{ A}$$

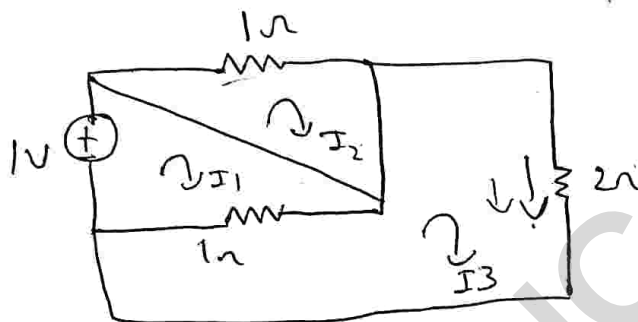
$$I = 0.4 + 0.25$$

$$I = 0.65 \text{ A}$$

7) Find the current through 2Ω using Superposition theorem



case 1) consider 1V



apply KVL to loop ①

$$-1I_1 + 1I_3 = -1 \quad \dots \textcircled{1}$$

apply KVL to loop ②

$$-1I_2 = 0 \quad \dots \textcircled{2}$$

apply KVL to loop 3

$$-2I_3 - (I_3 - I_1) = 0$$

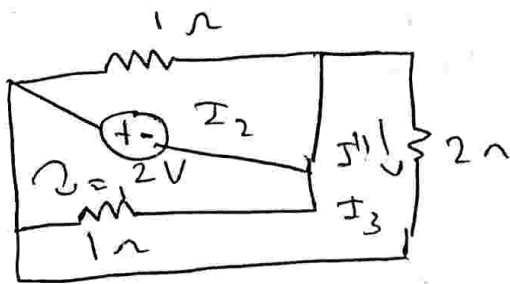
$$-2I_3 - I_3 + I_1 = 0$$

$$I_1 - 3I_3 = 0 \quad \dots \textcircled{3}$$

$$I_1 = 1.5A, I_2 = 0, I_3 = 0.5A$$

$$I_3 = I_1 = 0.5A$$

case (ii) consider 2V alone



loop ① KVL

$$-1I_1 + 1I_3 = 2 \quad \text{--- (1)}$$

loop ② KVL

$$-1I_2 = -2 \quad \text{--- (2)}$$

loop ③ KVL

$$-2I_3 - I_3 + I_1 = 0$$

$$I_1 - 3I_3 = 0 \quad \text{--- (3)}$$

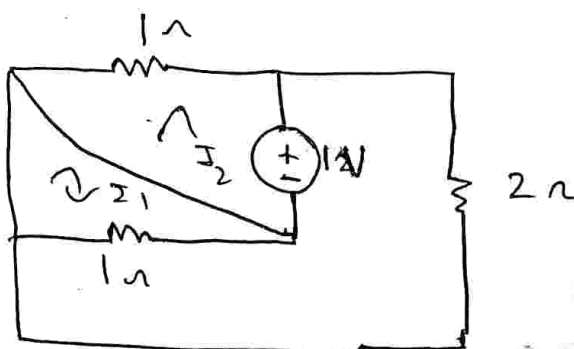
$$I_1 = -3$$

$$I_2 = 2$$

$$I_3 = -1$$

$$I_3 = I'' = -1A$$

case (iii) consider 1V alone



apply KVL to loop ①

$$-1I_1 + 1I_3 = 0$$

apply KVL to loop ③

$$I_1 - 3I_3 = -1V$$

apply KVL to loop 2

$$-1I_2 = 1V$$

$$I_1 = 0.5 \text{ A}$$

$$I_2 = \cancel{1} \text{ A}$$

$$I_3 = 0.5 \text{ A}$$

$$I_3 = I'' = 0.5 \text{ A}$$

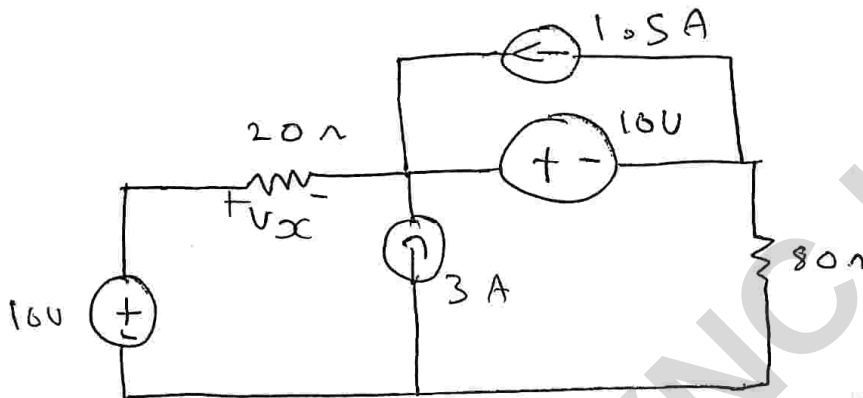
according to super position theorem

$$I = I' + I'' + I'''$$

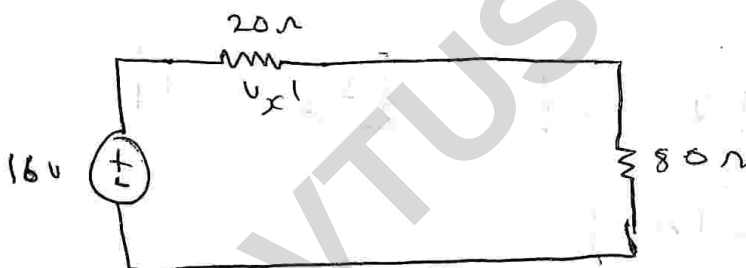
$$I = 0.5 - 1 + 0.5$$

$$I = 0 \text{ A}$$

8) for the ckt shown find V_x



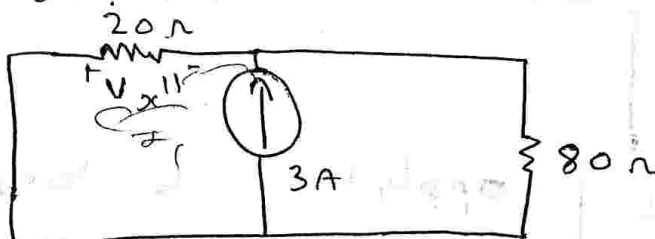
consider 16 V



$$V_{x1} = \frac{16 \times 20}{20 + 80}$$

$$V_{x1} = 3.2 \text{ V}$$

consider 3 A source.



apply current div.

$$I_{20} = \frac{3 \times 80}{20 + 80}$$

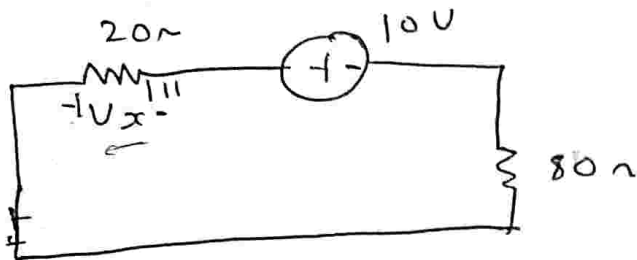
$$I_{20} = 2.4 \text{ A}$$

$$V_x^{II} = I_{10} \times R$$

$$V_x^{II} = 2.4 \times 20$$

$$V_x^{II} = -48V$$

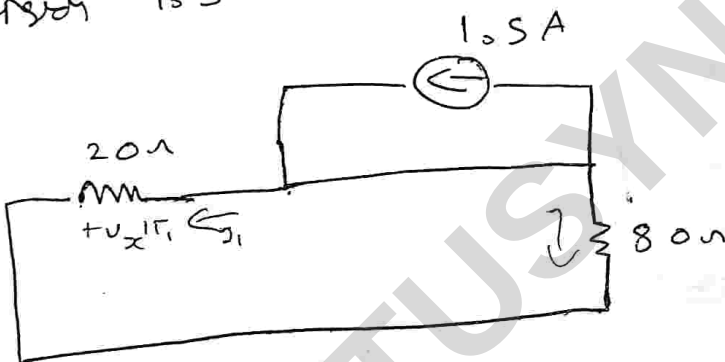
consider 10V alone.



$$V_x^{III} = \frac{10 \times 20}{80 + 20}$$

$$V_x^{III} = -2V$$

consider 1.5A alone



as there is no resistor with current hence the $I = 0$

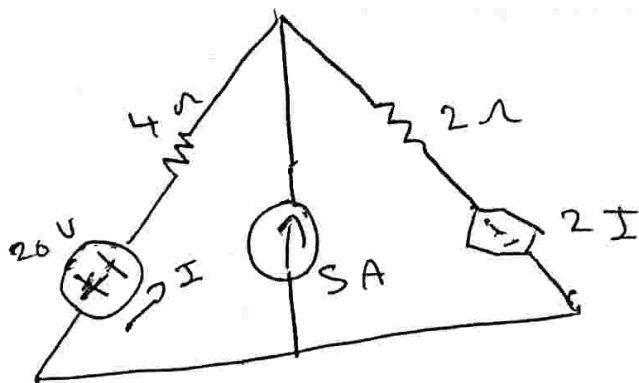
$$V_x^{IV} = 0V$$

$$\therefore V_x = V_x^I + V_x^{II} + V_x^{III} + V_x^{IV}$$

$$V_x = 3.2 - 2 + (-48)$$

$$V_x = -46.8V$$

9)

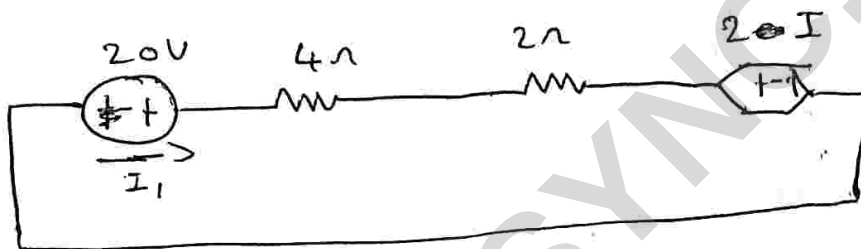


$$I_1 = 2 \text{ A}$$

$$I_2 = -1.25 \text{ A}$$

(In SPT if we have any dependent sources retain as it is in that)

consider 20V acting alone.



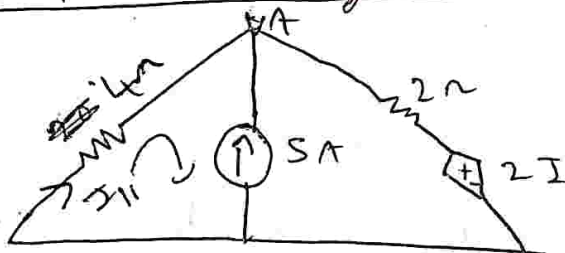
apply KVL

$$20 - 4I - I_2 - 2I = 0$$

$$20 - 8I = 0$$

$$I' = \frac{20}{8} = 2.5 \text{ A}$$

consider 5A acting alone:



Apply KVL at loop ①

$$-4I_{II} - 5 = 0$$

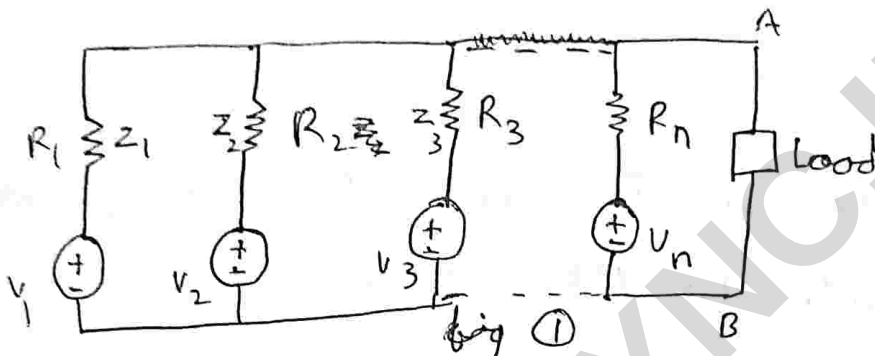
$$-4I_{II} = 5$$

$$I_{II} = -\frac{5}{4}$$

$$I_{II} = -1.25 \text{ A}$$

→ Millman's Theorem:

consider the following network



$Z = \text{impedance } (\Omega) \text{ ohm}$

$Y = \frac{1}{Z} \text{ Admittance}$

In any linear Bilateral network consisting of n number of voltage sources are converted into single voltage sources and all the impedances are converted into single impedance across the load terminal.

The single converted voltage source is given by

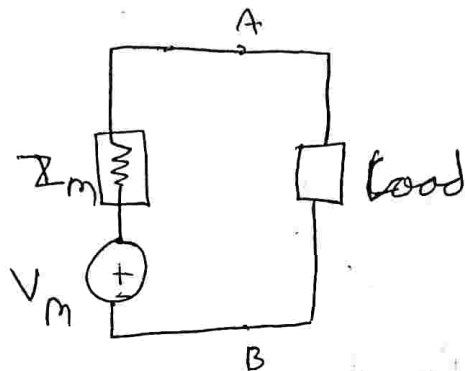
$$V_m = \frac{V_1 Y_1 + V_2 Y_2 + V_3 Y_3 + \dots + V_n Y_n}{Y_1 + Y_2 + Y_3 + \dots + Y_n}$$

$$V_m = \frac{\sum_{k=1}^n V_k Y_k}{\sum Y_k}$$

$$Z_m = \frac{1}{Y_1 + Y_2 + Y_3 \dots + Y_n}$$

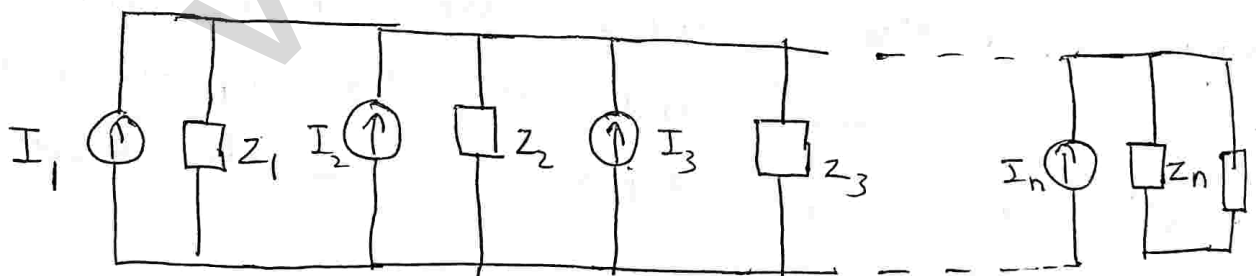
$$Z_m = \frac{1}{\sum_{K=1}^n Y_K}$$

millman's equivalent circuit is:



Proof: Consider the fig ① consists of n no of voltage sources connected in parallel across the load terminals $A B$

Convert all the voltage source into equivalent current sources



$$I_1 = \frac{V_1}{Z_1} = V_1 Y_1$$

$$I_2 = \frac{V_2}{Z_2} = V_2 Y_2$$

$$I_3 = \frac{V_3}{Z_3} = V_3 Y_3$$

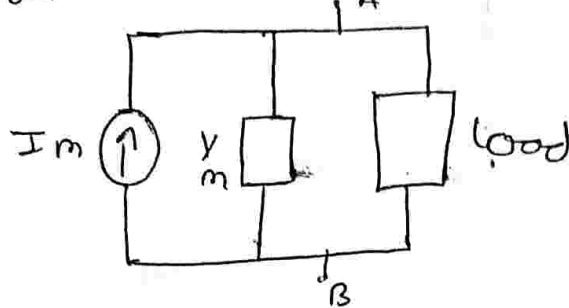
$$I_n = \frac{V_n}{Z_n} = V_n Y_n$$

We know that current sources and admittances in parallel can be added.

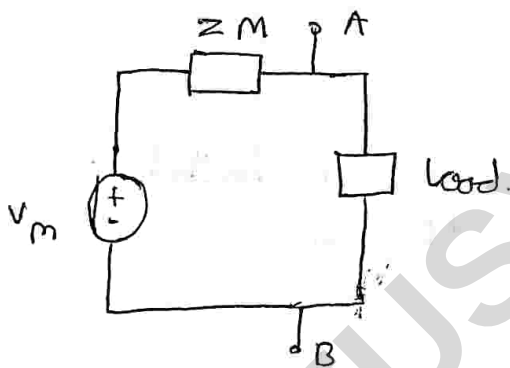
$$I_1 + I_2 + I_3 \dots \dots \dots I_n = I_m$$

$$Y_1 + Y_2 + Y_3 \dots \dots \dots Y_n = Y_m$$

The equivalent circuit is



convert above current source into equivalent voltage source.



$$V = IR$$

$$V_m = I_m Z_m$$

$$V_m = \frac{I_m}{Y_m} \dots \dots \textcircled{1}$$

$$V_m = \frac{I_1 + I_2 + I_3 \dots \dots + I_n}{Y_1 + Y_2 + Y_3 \dots \dots + Y_n} \dots \textcircled{2}$$

WKT

$$V = IZ$$

$$I = \frac{V}{Z} = VY \dots \textcircled{3}$$

Soln. B in (2)

$$V_m = \frac{V_1 Y_1 + V_2 Y_2 + V_3 Y_3 + \dots + V_n Y_n}{Y_1 + Y_2 + Y_3 + \dots + Y_n}$$

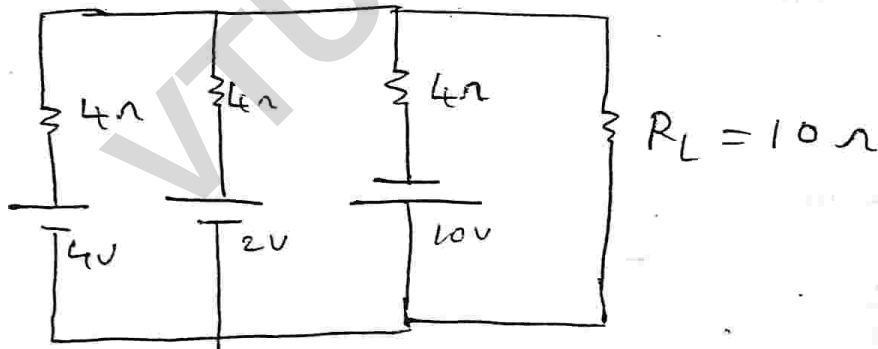
Tejashree S, Dept of ECE, SVIT

$$V_m = \frac{\sum_{K=1}^n V_K Y_K}{\sum_{K=1}^n Y_K}$$

$$Z_m = \frac{1}{Y_1 + Y_2 + Y_3 + \dots + Y_n}$$

$$Z_m = \frac{1}{\sum_{K=1}^n Y_K}$$

① Find the value of current in ~~load~~ resistor $R_L = 10 \Omega$ using millman's theorem



Soln. $Z_1 = 4 \quad Z_2 = 4 \quad Z_3 = 4$

$$Y_1 = Y_2 = Y_3 = \frac{1}{4} = 0.25$$

$$V_1 = 4 \quad V_2 = 2 \quad V_3 = -10$$

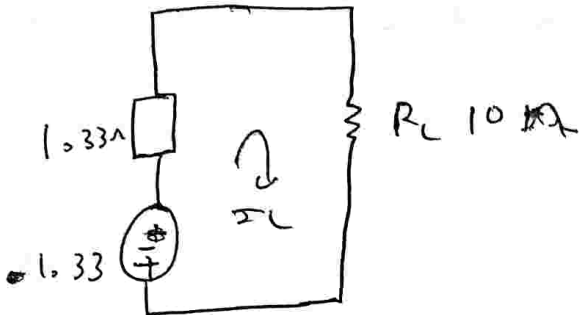
$$V_m = \frac{V_1 Y_1 + V_2 Y_2 + V_3 Y_3}{Y_1 + Y_2 + Y_3}$$

$$V_m = \frac{(4 \times 0.25) + (2 \times 0.25) + (-10 \times 0.25)}{0.75}$$

$$V_m = -1.33 \text{ V}$$

$$Z_m = \frac{1}{0.75}$$

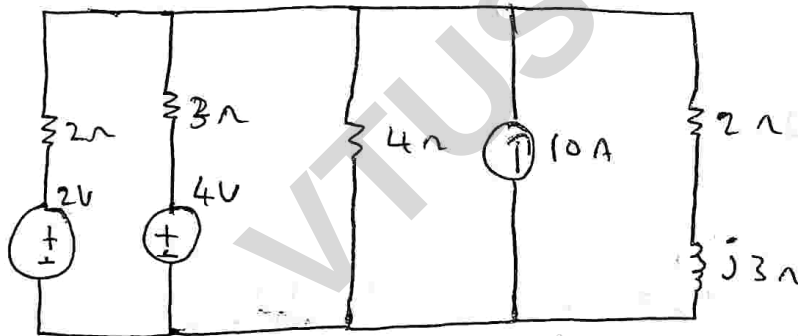
$$Z_m = 1.33 \Omega$$



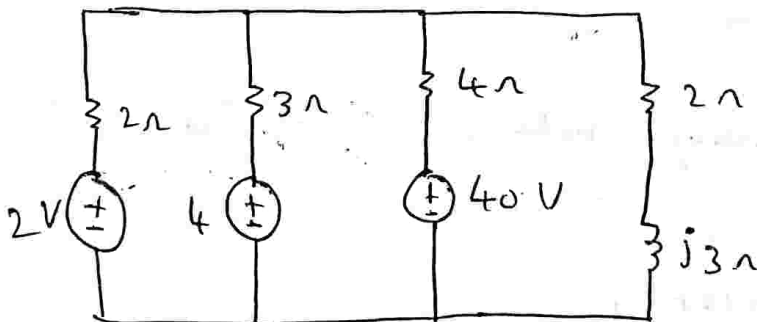
$$I_L = \frac{1.33}{1.33 \times 10}$$

$$I_L = 0.117 \text{ A}$$

②



Find the current $Z_L = 2 + j3 \Omega$



$$Z_1 = 2 \quad Z_2 = 3 \quad Z_3 = 4$$

$$Y_1 = \frac{1}{2} = 0.5 \quad Y_2 = \frac{1}{3} = 0.33 \quad Y_3 = \frac{1}{4} = 0.25$$

$$V_1 = 2V \quad V_2 = 4V \quad V_3 = 40V$$

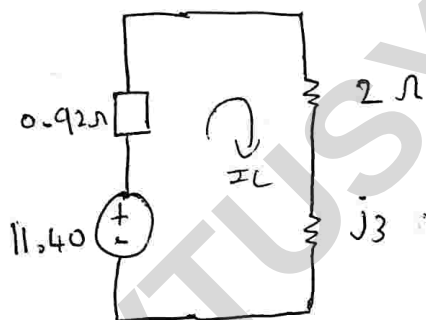
$$V_m = \frac{V_1 X_1 + V_2 X_2 + V_3 X_3}{X_1 + X_2 + X_3}$$

$$V_m = \frac{(2 \times 0.5) + (4 \times 0.33) + (0.25 \times 40)}{0.5 + 0.33 + 0.25}$$

$$V_m = 11.40V$$

$$Z_m = \frac{1}{X_1 + X_2 + X_3} = \frac{1}{0.5 + 0.33 + 0.25}$$

$$Z_m = 0.92 \Omega$$



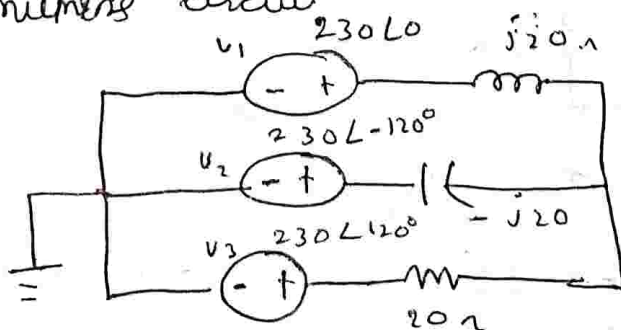
$$I_L = \frac{V}{R}$$

$$I_L = 1.899 - 1.95jA$$

or

$$I_L = 2.72 \angle -45.7^\circ A$$

③. convert the following network into equivalent circuit



$$Z_1 = j20$$

$$Z_2 = -j20$$

$$Z_3 = 20$$

$$Y_1 = \frac{1}{j20} = -0.05j$$

$$Y_2 = \frac{1}{-j20} = 0.05j$$

$$Y_3 = \frac{1}{20}$$

$$Y_3 = 0.05$$

$$V_1 = 230 \angle 0^\circ = 230$$

$$V_2 = 230 \angle -120^\circ$$

$$V_2 = -115 - 199.18j$$

$$V_3 = 20 \angle 120^\circ$$

$$V_3 = -115 + 199.18j$$

$$Z_m = \frac{1}{Y_1 + Y_2 + Y_3}$$

$$Z_m = \frac{1}{-0.05j + 0.05j + 0.05}$$

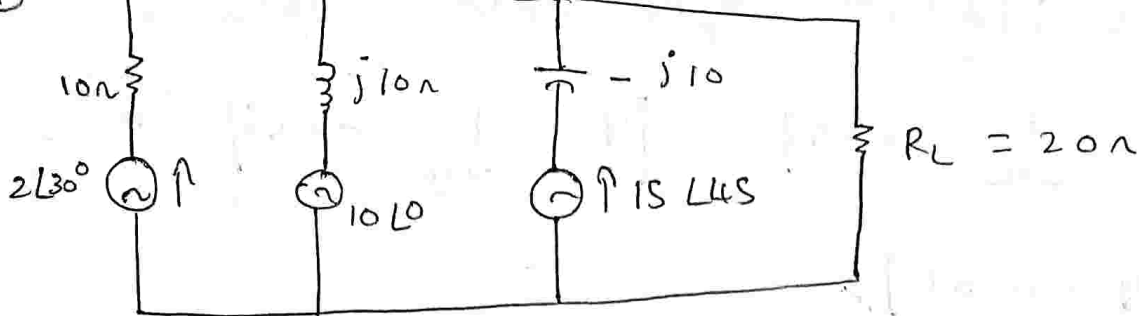
$$Z_m = 20 \Omega$$

$$V_m = \frac{V_1 Y_1 + V_2 Y_2 + V_3 Y_3}{Y_1 + Y_2 + Y_3}$$

$$V_m = \frac{230(-0.05j) + (-115 - 199.18j)(0.05j) + (-115 + 199.18j)(0.05)}{-0.05j + 0.05j + 0.05}$$

$$V_m = 84.18 - 145.82j \text{ V}$$

$$V_m = 168.37 \angle 60^\circ \text{ V}$$



$$Z_1 = 10$$

$$Z_2 = j10$$

$$Z_3 = -j10$$

$$Y_1 = \frac{1}{Z_1} = 0.1$$

$$Y_2 = \frac{1}{j10} = -0.01j$$

$$Y_3 = \frac{1}{-j10} = 0.01j$$

$$V_1 = 2 \angle 30^\circ \quad V_1 = 1.73 + j1$$

$$V_2 = 10$$

$$V_3 = 15 \angle 45^\circ \quad V_3 = 10.60 + j10.60$$

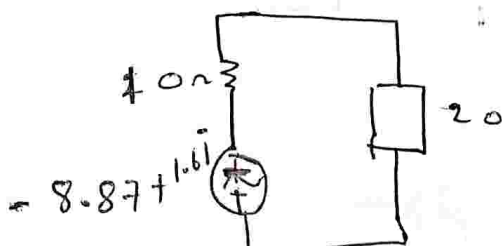
$$Z_m = \frac{1}{0.1 + (-0.01j) + (0.01j)}$$

$$Z_m = 10 \Omega$$

$$V_m = \frac{V_1 Y_1 + V_2 Y_2 + V_3 Y_3}{Y_1 + Y_2 + Y_3}$$

$$V_m = \frac{(1.73 + j1)(0.1) + (10)(-0.01j) + (10.60 + j10.60)(0.01j)}{0.1 + (-0.01j) + (0.01j)}$$

$$V_m = -8.87 + j1.61 \text{ V}$$

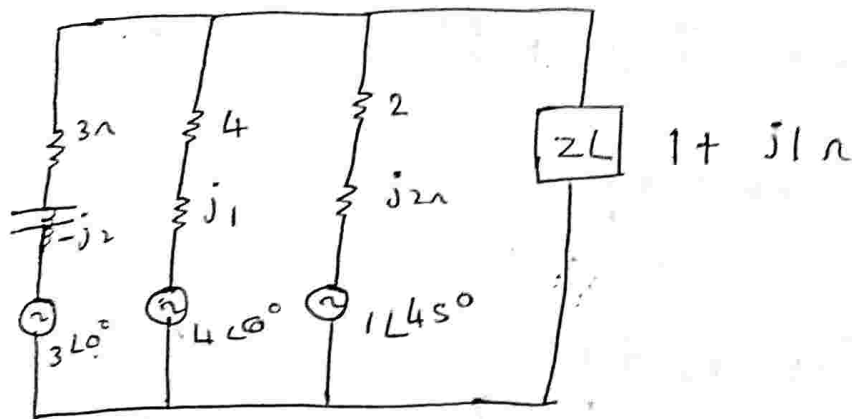


$$I_L = 0.3 \angle 169.77^\circ \text{ A}$$

$$I_L = \frac{-8.87 + j1.61}{20 + j10}$$

$$I_L = -0.295 + j0.051 \text{ A}$$

5. Find the current through $1 + j1 \Omega$



$$Z_1 = 3 - j2$$

$$Z_2 = 4 + j1$$

$$Z_3 = 2 + j2 \Omega$$

$$Y_1 = \frac{1}{Z_1} = \frac{1}{3 - j2}$$

$$Y_2 = \frac{1}{4 + j1}$$

$$Y_3 = \frac{1}{2 + j2}$$

$$Y_1 = 0.23 + 0.15j$$

$$Y_2 = 0.23 - 0.05j$$

$$Y_3 = 0.25 - 0.25j$$

$$Z_m = \frac{1}{Y_1 + Y_2 + Y_3}$$

$$Z_m = \frac{1}{(0.23 + 0.15j) + (0.23 - 0.05j) + (0.25 - 0.25j)}$$

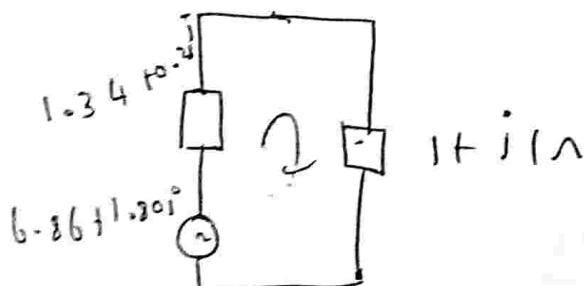
$$Z_m = 1.34 + 0.28j$$

$$V_1 = 3$$

$$V_2 = 4$$

$$V_3 = 0.707 + 0.707j$$

$$V_m = 6.86 + 1.80j \text{ V}$$



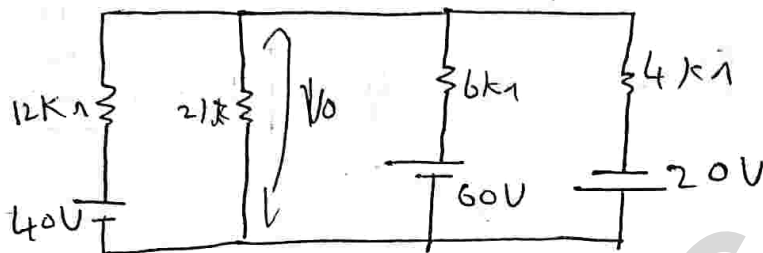
$$\underline{I}_L = \frac{6.86 + 1.80j}{(1.34 + 0.28j) + (1 + 1j)}$$

$$\underline{I}_L = 2.58 - 0.64j \text{ A}$$

or

$$\underline{I}_L = 2.65 \angle -13.97^\circ \text{ A}$$

6)



$$Z_1 = 12k\Omega$$

$$Z_2 = 6k$$

$$Z_3 = 4k$$

$$y_1 = \frac{1}{12 \times 10^3} = 8.33 \times 10^{-5}$$

$$y_2 = \frac{1}{6 \times 10^3} = 1.66 \times 10^{-4}$$

$$y_3 = \frac{1}{4k} = 2.5 \times 10^{-4}$$

$$Z_m = \frac{1}{y_1 + y_2 + y_3}$$

$$Z_m = \frac{1}{8.33 \times 10^{-5} + 2.5 \times 10^{-4} + 1.66 \times 10^{-4}}$$

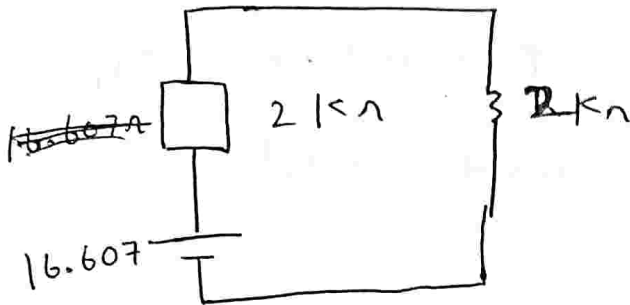
$$Z_m = 2k\Omega$$

$$V_m = \frac{V_1 y_1 + V_2 y_2 + V_3 y_3}{y_1 + y_2 + y_3}$$

$$V_m = (40 \times 8.33 \times 10^{-5}) + (60 \times 1.66 \times 10^{-4}) + (-20 \times 2.5 \times 10^{-4})$$

$$8.33 \times 10^{-5} + 1.66 \times 10^{-4} + 2.5 \times 10^{-4}$$

$$V_m = 16.607 \text{ V}$$



$$V_o = \frac{16.607 \times 2 \times 10^3}{2 \times 10^3 + 2 \times 10^3}$$

$$V_o = 8.3 \text{ V}$$

Thevenin's Theorem:

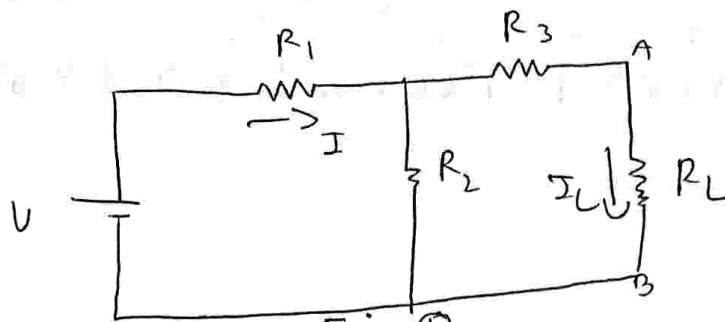
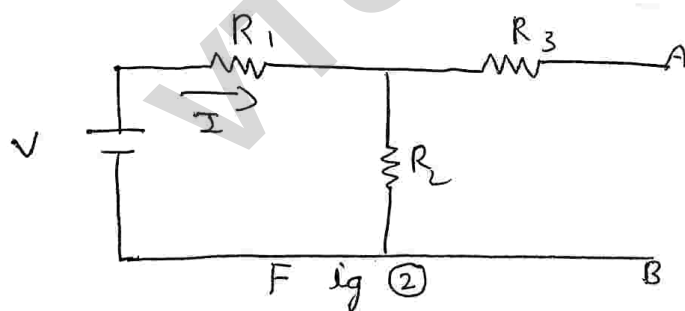


Fig ①

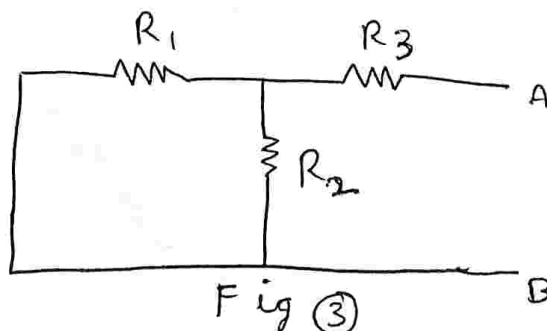
In any linear Bilateral network is represented by a single voltage ~~source~~ V_{th} & Resistance R_{th} and current through the load resistance is given by.

$$I_L = \frac{V_{th}}{R_{th} + R_L}$$

Thevenin's Voltage is obtained by load resistance is removed & R_{th} is obtained by removing the load as well as if there is any voltage source make it short ckt, if any current source open ckt (open the current).



$$V_{th} = I \cdot R_2$$



$$R_{th} = (R_1 \parallel R_2) + R_3$$

And consider the n/w shown in fig ①

from ckt

$$I_L = \frac{I R_2}{R_2 + (R_3 + R_L)} \dots \textcircled{1}$$

$$I = \frac{V}{(R_3 + R_L \parallel R_2) + R_1}$$

$$I = \frac{V}{R_1 + \frac{R_2 (R_3 + R_L)}{R_2 + R_3 + R_L}}$$

$$I = \frac{V}{\frac{R_1 R_2 + R_1 R_3 + R_1 R_L + R_2 R_3 + R_3 R_L}{R_2 + R_3 + R_L}}$$

$$I = \frac{V (R_2 + R_3 + R_L)}{R_1 R_2 + R_1 R_3 + R_1 R_L + R_2 R_3 + R_3 R_L} \dots \textcircled{2}$$

sub ② in ①

$$I_L = \frac{V (\cancel{R_2} + \cancel{R_3} + \cancel{R_L})}{R_1 R_2 + R_1 R_3 + R_1 R_L + R_2 R_3 + R_3 R_L} \cdot R_2$$
$$\frac{\cancel{R_2} + (\cancel{R_3} + \cancel{R_L})}{R_1 R_2 + R_1 R_3 + R_1 R_L + R_2 R_3 + R_3 R_L}$$

$$I_L = \frac{V R_2}{R_1 R_2 + R_1 R_3 + R_1 R_L + R_2 R_3 + R_3 R_L} \dots \textcircled{3}$$

To prove:

To find V_{th} ,

consider fig ②

$$V_{th} = I R_2 \quad \dots \quad (4)$$

To find R_{th}

consider fig ③

$$R_{th} = (R_1 \parallel R_2) + R_3$$

$$R_{th} = \frac{R_1 R_2}{R_1 + R_2} + R_3$$

$$R_{th} = \frac{R_1 R_2 + R_1 R_3 + R_2 R_3}{R_1 + R_2} \quad \dots \quad (5)$$

The current through load resistance is given by.

$$I_L = \frac{V_{th}}{R_{th} + R_L} \quad \dots \quad (6)$$

$$I_L = \frac{I R_2}{$$

$$\left(\frac{R_1 R_2 + R_1 R_3 + R_2 R_3}{R_1 + R_2} \right) + R_L$$

$$I_L = \frac{I R_2 (R_1 + R_2)}{$$

$$R_1 R_2 + R_1 R_3 + R_2 R_3 + R_1 R_L + R_2 R_L$$

$$I = \frac{V}{R_1 + R_2} \dots \textcircled{7}$$

$$I_L = \frac{V}{R_1 + R_2} R_2 \quad (\cancel{R_1 + R_2})$$

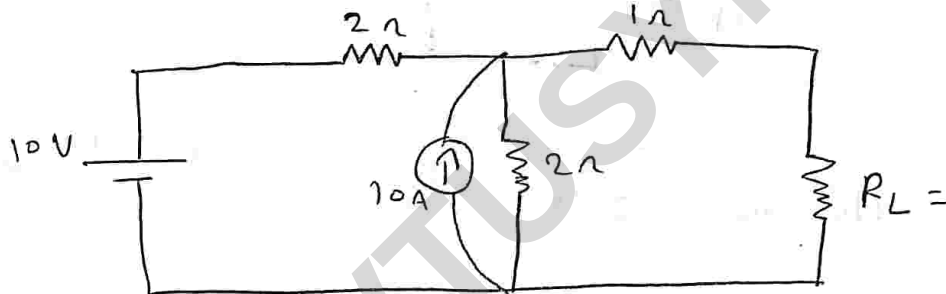
$$R_1 R_2 + R_1 R_3 + R_2 R_3 + R_1 R_L + R_2 R_L$$

$$I_L = \frac{V R_2}{R_1 R_2 + R_1 R_3 + R_2 R_3 + R_1 R_L + R_2 R_L} \dots \textcircled{8}$$

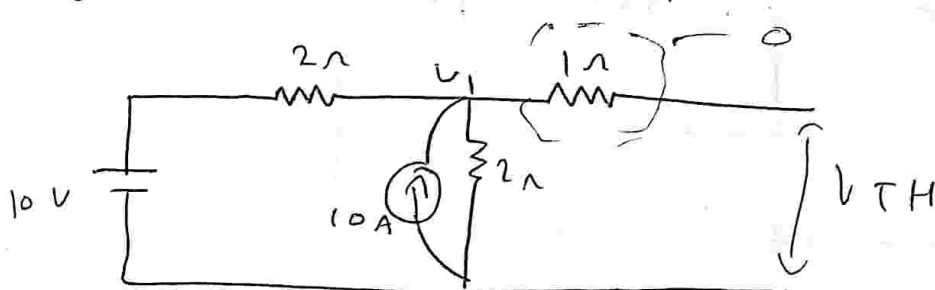
$$\text{eq } \textcircled{3} = \text{eq } \textcircled{8}$$

Hence proved //

1) Using Thevenin's theorem Find the load current



To find V_{TH}



$$V_{TH} = V_1$$

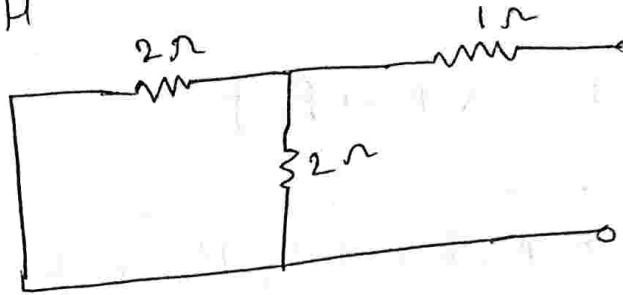
$$\frac{V_1 - 10}{2} + \frac{V_1}{2} = 10$$

$$\frac{V_1}{2} + \frac{V_1}{2} = 10 + 5$$

$$V_1 = 15$$

$$V_{TH} = 15V$$

To find R_{TH}



$$R_{TH} = (2 || 2) + 1$$

$$R_{TH} = 2 \Omega$$

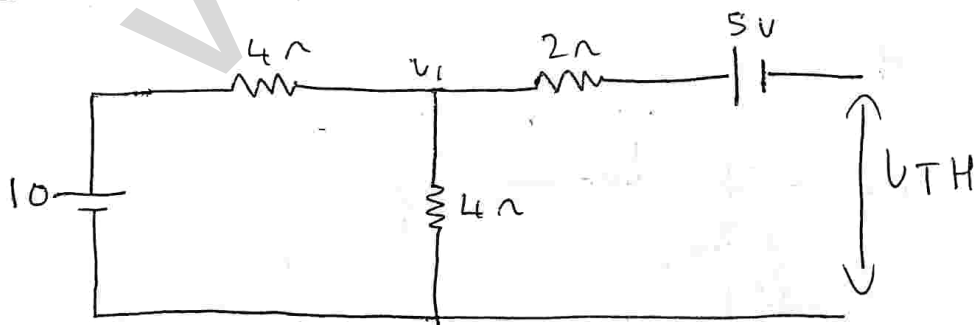
$$I_L = \frac{V_{TH}}{R_{TH} + R_L}$$

$$I_L = \frac{15}{2 + 1}$$

$$I_L = 5A$$



2) using thevenin theorem find the current through R_L



$$\frac{V_1 - 10}{4} + \frac{V_1}{4} + \frac{V_1}{2} = 0$$

$$V_{TH} = V_1 - 5$$

$$V_1 = 15$$

$$V_{TH} = 15 - 5$$

$$\frac{V_1}{4} - \frac{10}{4} +$$

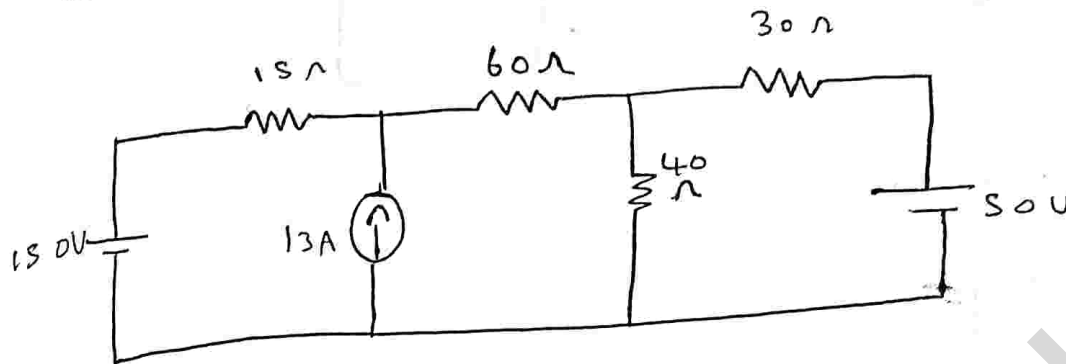
$$\frac{V_1}{2} = 10 - 5$$

$$V_{TH} = 10V$$

$$R_{Th} = (2 || 2) + 2$$

$$R_{Th} = 4\Omega$$

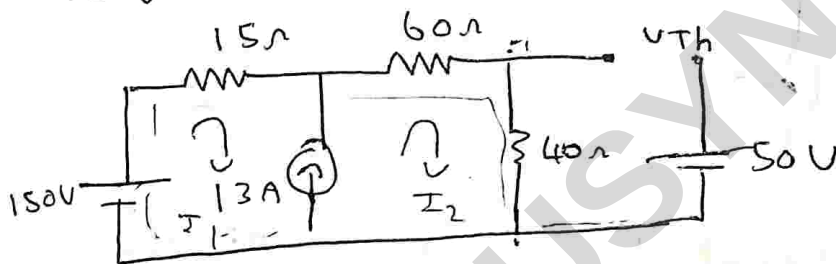
3) using thevenin theorem find the current through 30Ω resistor.



Soln:

$$R_L = 30\Omega$$

To find R_{Th} open $R_L = 30\Omega$



$$I_2 - I_1 = 13 \quad \text{--- (1)}$$

apply KVL to super mesh

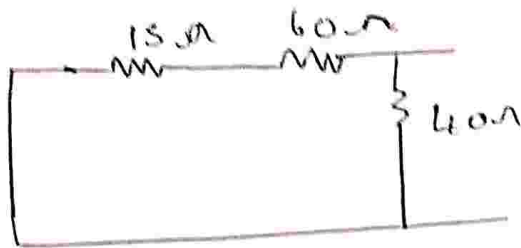
$$150 - 15I_1 - 60I_2 - 40I_2 = 0$$

$$-15I_1 - 100I_2 = -150$$

$$I_1 = -10A$$

$$I_2 = 3$$

$$V_{Th} = 70V$$

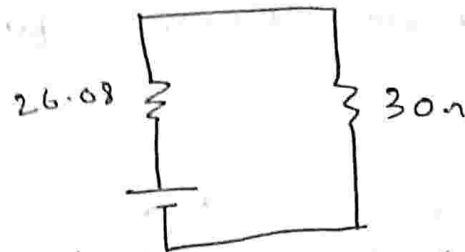


$$R_{TH} = 26.08 \Omega$$

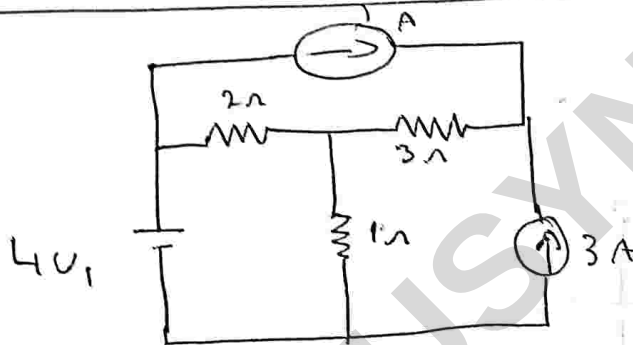
$$I_L = \frac{V_{TH}}{R_{TH} + R_L}$$

$$I_L = \frac{70}{26.08 + 30}$$

$$I_L = 1.24 A$$

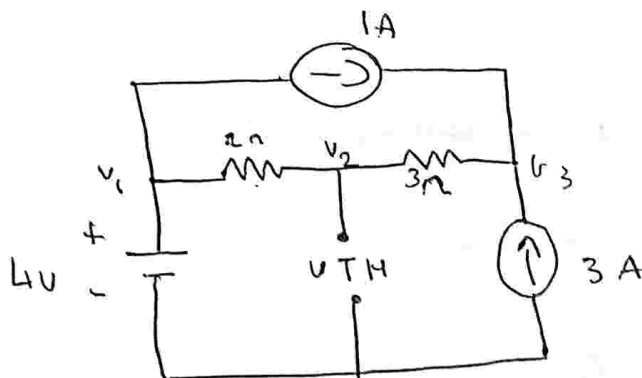


4)



Find the current through 1Ω using thevenin's theorem

soln: Remove R_L 1Ω



From the ckt $V_1 = 4V$

$$V_2 = V_{TH}$$

$$\frac{V_2 - 4}{2} + \frac{V_2 - V_3}{3} = 0$$

$$V_2 \left(\frac{1}{2} + \frac{1}{3} \right) - V_3 \left(\frac{1}{3} \right) = 2$$

$$0.833 V_2 - 0.333 V_3 = 2$$

apply KCL to node ②

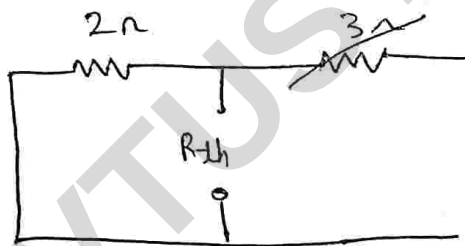
$$\frac{V_3 - V_2}{3} = 1 + 3$$

$$-0.333 V_2 + 0.333 V_3 = 4$$

$$V_2 = 12 V$$

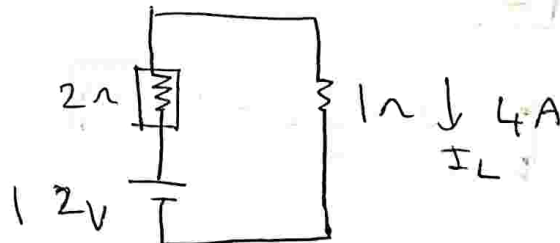
$$\therefore V_{TH} = 12 V$$

$$R_{TH} =$$



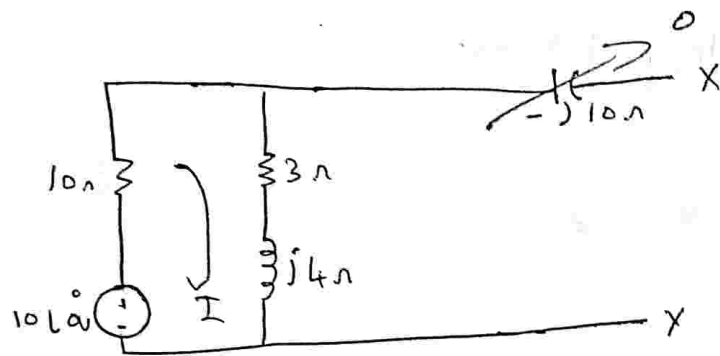
$$R_{TH} = 2 \Omega$$

$$I_L = \frac{12}{2+1}$$



$$I_L = 4 A$$

5) obtain the thevenin equivalent for the ckt shown across the terminal x and y



To find V_{TH} open load.

$$V_{TH} = I(3 + j4)$$

apply KVL to loop 1

$$10 - 10I - (3 + j4)I = 0$$

$$10(-13 - 4j)I = 0$$

$$I = \frac{-10}{-13 - 4j}$$

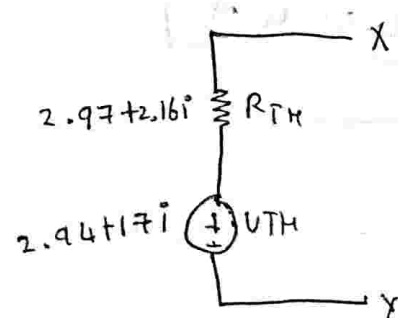
$$I = 0.70 - 0.21j$$

$$I = 0.73 \angle -17.10^\circ$$

$$V_{TH} = 0.70 - 0.21j(3 + j4)$$

$$V_{TH} = 3.654 \angle 36.43^\circ$$

$$V_{TH} = 2.94 + 2.17j$$



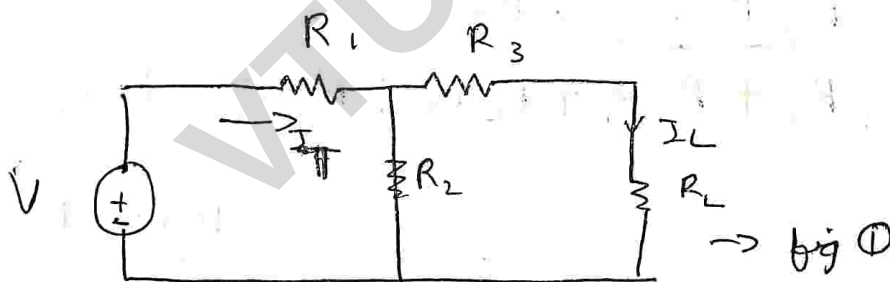
$$R_{TH} = 10 \parallel (3 + j4)$$

$$R_{TH} = 2.97 + 2.16j$$

→ Norton's theorem:

In any linear bilateral network, however the network may be complicated, the N/W may be replaced by single current source in parallel with an impedance.

The current of current source is equal to the short circuit current flowing through the load terminal and the value of Z is equal to the impedance of the N/W ~~seen~~ viewed from the load terminal and replace all independent voltage source by short ckt and current by open ckt.



consider the following ckt
from the ckt

$$I_L = \frac{I_T R_2}{R_2 + R_L + R_3} \quad \text{--- (1)}$$

$$I_T = \frac{V}{R_T} \quad \text{--- (2)}$$

$$R_T = (R_3 + R_L \parallel R_2) + R_1$$

$$R_T = R_1 + \frac{R_2 (R_3 + R_L)}{R_2 + R_3 + R_L}$$

$$R_T = \frac{R_1 R_2 + R_3 R_1 + R_L R_1 + R_2 R_3 + R_2 R_L}{R_2 + R_3 + R_L}$$

subs ③ in ②

$$I_T = \frac{V (R_2 + R_3 + R_L)}{R_1 R_2 + R_3 R_1 + R_L R_1 + R_2 R_3 + R_2 R_L} \quad \text{--- ④}$$

subs 4 in ①

$$I_L = \frac{V \cancel{R_2} + \cancel{R_3} + \cancel{R_L}}{R_1 \cancel{R_2} + R_3 R_1 + R_L R_1 + R_2 R_3 + R_2 R_L} \times R_2$$

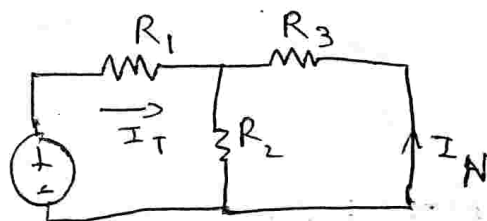
$$\cancel{R_3} + \cancel{R_2} + \cancel{R_L}$$

$$I_L = \frac{V R_2}{R_1 R_2 + R_3 R_1 + R_L R_1 + R_2 R_3 + R_2 R_L}$$

neglect R_3

$$I_L = \frac{V R_2}{R_1 R_2 + R_L R_1 + R_2 R_L}$$

case ① To find I_L short R_L



applying ~~current~~ current divider

$$I_N = \frac{I_T \cdot R_2}{R_2 + R_3} \dots \textcircled{5}$$

$$I_N = \frac{V}{R_T} \dots \textcircled{6}$$

$$R_T = R_1 + (R_2 \parallel R_3)$$

$$R_T = R_1 + \frac{(R_2 \parallel R_3)}{R_2 \parallel R_3}$$

$$R_T = \frac{R_1 R_2 + R_1 R_3 + R_2 R_3}{R_2 + R_3} \dots \textcircled{7}$$

subs $\textcircled{7}$ in $\textcircled{6}$

$$I_T = \frac{V (R_2 + R_3)}{R_1 R_2 + R_1 R_3 + R_2 R_3} \dots \textcircled{8}$$

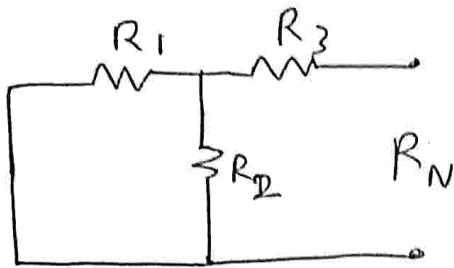
subs $\textcircled{8}$ in $\textcircled{5}$

$$I_N = \frac{V (R_2 + R_3)}{R_1 R_2 + R_1 R_3 + R_2 R_3} \cdot \frac{R_2}{R_2 + R_3}$$

$$\boxed{I_N = \frac{V R_2}{R_1 R_2 + R_1 R_3 + R_2 R_3}} \dots \textcircled{9}$$

To find R_N

Voltage source shorted, current open, R_{load} removed.



$$R_N = (R_1 \parallel R_2) + R_3$$

$$R_N = \frac{R_1 R_2}{R_1 + R_2} + R_3$$

$$R_N = \frac{R_1 R_2 + R_3 R_1 + R_2 R_3}{R_1 + R_2}$$

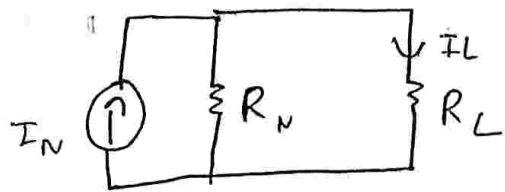
$$I_L = \frac{I_N \times R_N}{R_N + R_L}$$

$$I_L = \frac{V R_2}{(R_1 R_2 + R_1 R_3 + R_2 R_3)} \times \frac{(R_1 R_2 + R_3 R_1 + R_2 R_3)}{(R_1 + R_2)}$$

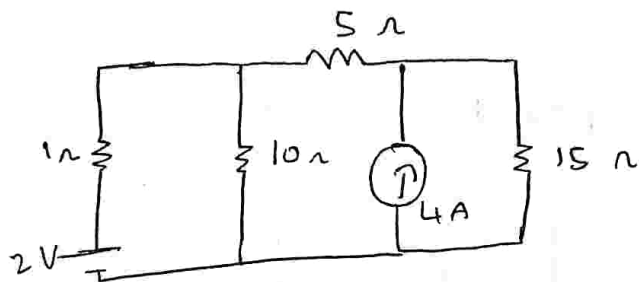
$$\frac{R_1 R_2 + (R_3 R_1 + R_2 R_3)}{(R_1 + R_2)} + R_L (R_1 + R_2)$$

$$I_L = \frac{V R_2}{R_1 R_2 + R_3 R_1 + R_L R_1 + R_2 R_3 + R_2 R_L}$$

The Norton equivalent circuit is



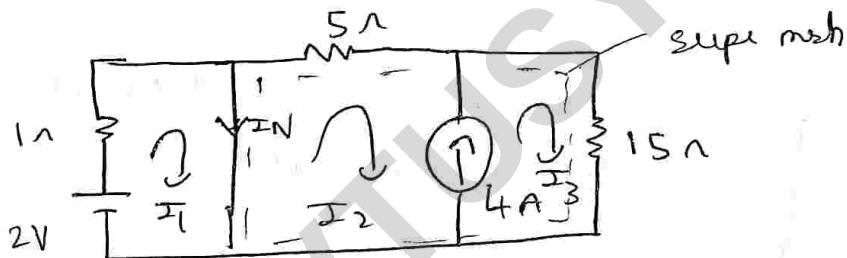
1) using Norton's theorem find the current through 10Ω resistance



soln.

$$R_L = 10\Omega$$

short 10Ω Replace by I_N



from C.R.A

$$I_N = I_1 - I_2 \quad \text{--- (1)}$$

$$I_3 - I_2 = 4 \quad \text{--- (2)}$$

apply KVL to super mesh.

$$-5I_2 - 15I_3 = 0 \quad \text{--- (3)}$$

$$I_2 = -3$$

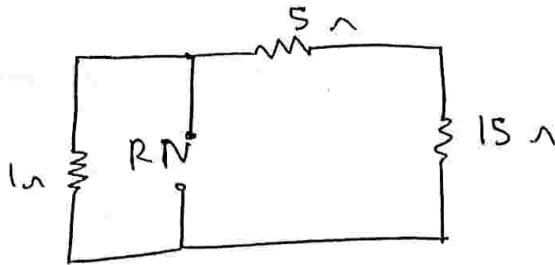
$$I_3 = 1$$

$$\therefore I_N = 2 - (-3)$$

$$I_N = 5A$$

$$I_1 = \frac{V}{R} = \frac{2}{1} = 2A$$

To find R_N



$$5 + 15 \parallel 1$$

$$R_N = \frac{20}{21}$$

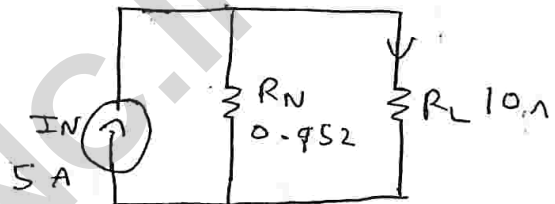
$$R_N = 0.952 \Omega$$

$$I_L = \frac{I_N R_N}{R_N + R_L}$$

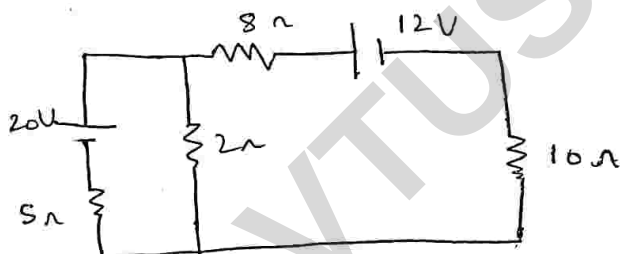
$$I_L = \frac{5 \times 0.952}{0.952 + 10}$$

$$I_L = 0.43 \text{ A}$$

Norton's equivalent circuit

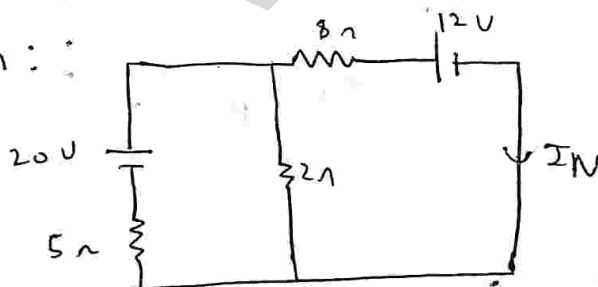


2)



Find the current through 10 ohm Resistor through R_L

Soln:



$$I_N = -0.666 \text{ A}$$

$$R_N = 9.429 \Omega$$

$$I_L = -0.326 \text{ A}$$

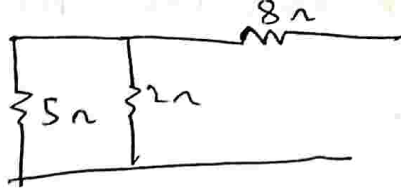
$$-5I_1 - 2I_1 + 2I_2 = -20 \quad \text{--- (1)}$$

$$-2I_2 + 2I_1 - 8I_2 = 12 \quad \text{--- (2)}$$

$$I_1 = 2.66 \quad I_2 = -0.666 \text{ A}$$

$$\therefore I_N = -0.666 \text{ A}$$

To $R_N =$



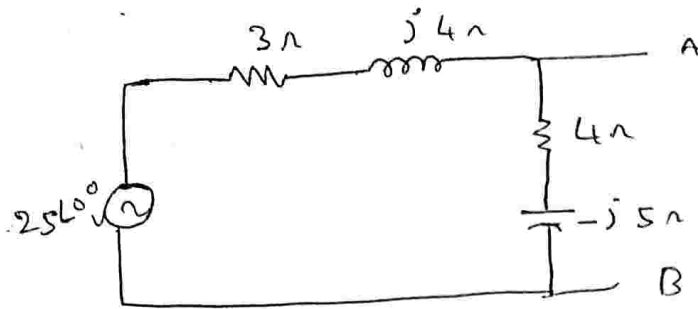
$$R_N = (5 \parallel 2) + 8$$

$$R_N = 9.428 \Omega$$

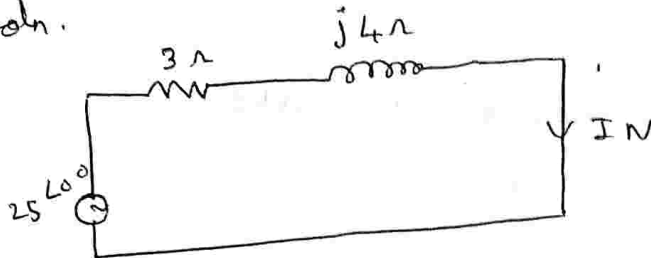
$$I_L = \frac{R_N \times I_N}{R_N + R_L} = \frac{9.428 \times (-0.666)}{9.428 + 10}$$

$$I_L = -0.3231 \text{ A}$$

3) obtain the Norton equivalent for the n/w shown



Soln.



$$-3I_1 - j4I_1 = 25$$

$$(-3 - 4j)I_1 = 25$$

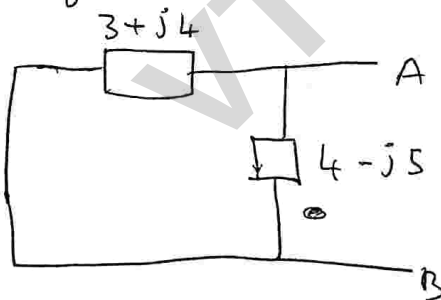
$$I_1 = \frac{25}{-3 - 4j}$$

$$I_N = -3 + 4j$$

$$I_N = 3 - 4j$$

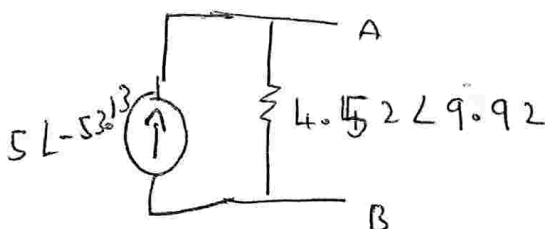
$$I_1 = 5 \angle -53.13^\circ$$

To find R_N

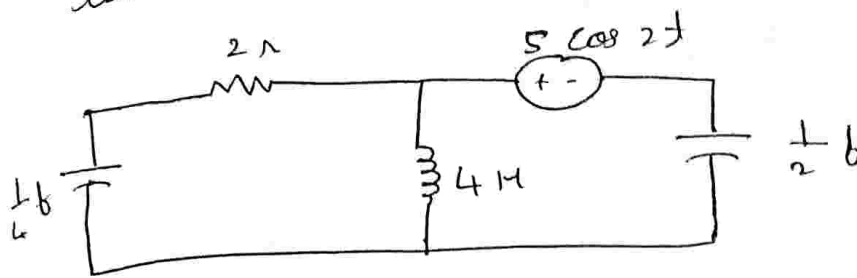


$$R_N = 4.52 \angle 9.92^\circ$$

$$R_N = 4.46 + 0.78j$$



4) Determine the current through 4 H for the ckt shown using Norton's theorem.



Represent the given ckt in frequency domain $\omega = 2$ Radians/sec.

$$X_L = j\omega L$$

$$X_L = j \times 2 \times 4$$

$$X_L = j8\Omega$$

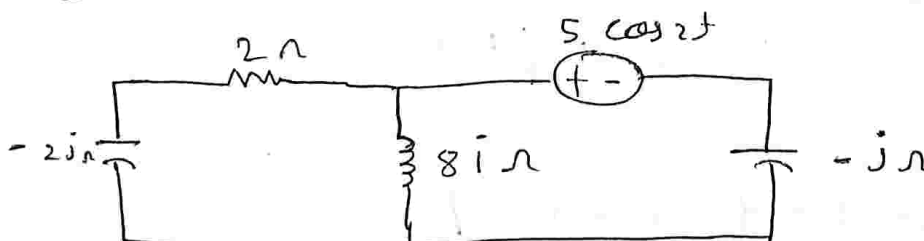
$$X_C = \frac{1}{j\omega C}$$

$$X_C = \frac{1}{j \times 2 \times 0.5}$$

$$X_C = -j\Omega$$

$$X_C = \frac{1}{j \times 2 \times 0.25}$$

$$X_C = -2j\Omega$$

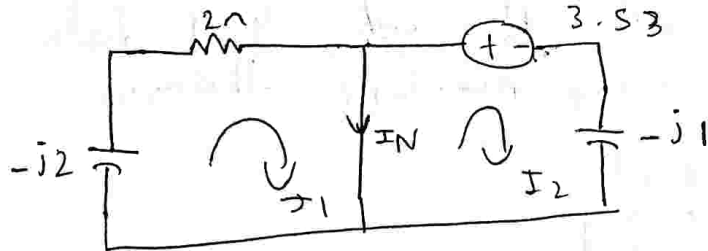


$5 \cos 2t$ is of the form $V = V_m \cos \omega t$

$$V_m = 5$$

$$\omega = 2$$

$$V_{rms} = \frac{V_m}{\sqrt{2}} = \frac{5}{\sqrt{2}} = 3.53 \text{ V}$$



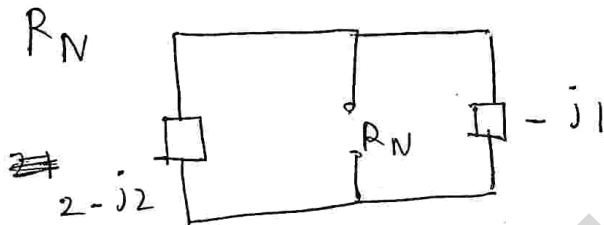
$$I_N = I_2 - I_1$$

$$I_1 = 0$$

$$I_2 = \frac{V}{R} = \frac{3.53}{-j1}$$

$$I_N = 3.53 i$$

$$\Rightarrow I_N = 3.53 \angle 90^\circ$$



$$R_N = \frac{(2 - j2) \times (-j1)}{(2 - j2) + (-j1)}$$

$$R_N = 0.153 - 0.76 i$$

$$R_N = 0.78 \angle -78.1^\circ$$

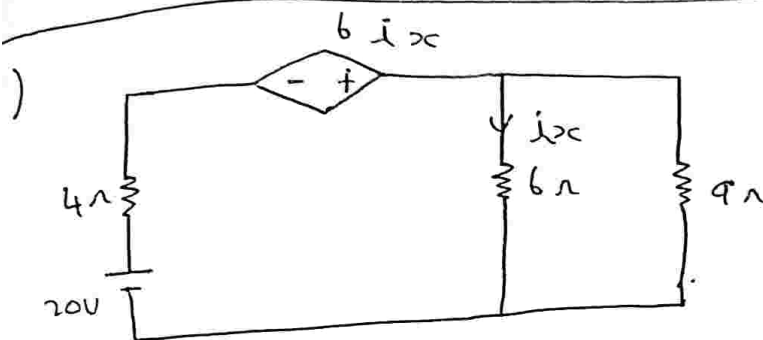
$$I_L = \frac{R_N \times I_N}{R_N + R_L}$$

$$I_L = \frac{(0.78 \angle -78.69^\circ) (3.53 \angle 90^\circ)}{(0.78 \angle -78.69^\circ) + (8 i)}$$

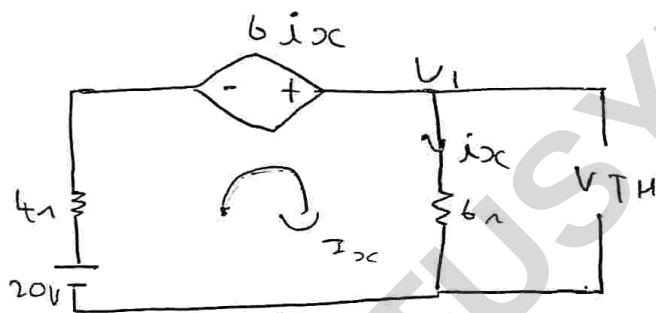
$$I_L = 0.380 \angle -77.47^\circ \text{ A}$$

→ Thevenin's & Norton's theorem for the ckt with dependent sources:

- Identify and remove the load resistance
- look into the load terminal & calculate V_{TH}
- place a short on load terminal calculate the I_N & determine $R_{TH} = \frac{V_{TH}}{I_N}$
- The load current $I_L = \frac{V_{TH}}{R_{TH} + R_L}$



Determine the current through 9Ω



from the ckt
 $V_{TH} = 6i_x$

$$20 - 4I_x + 6I_x - 6I_x = 0$$

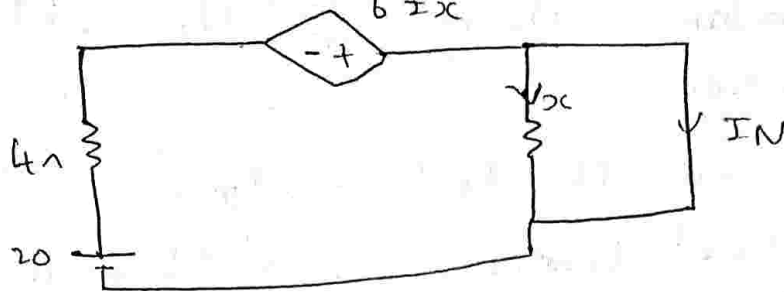
$$20 = 4I_x$$

$$I_x = \frac{20}{4}$$

$$I_x = 5A$$

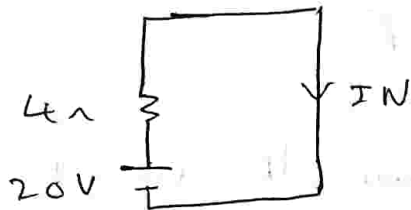
$$V_{TH} = 6 \times 5$$

$$V_{TH} = 30V$$



Since R is connected in || with short ckt here

$$I_{sc} = 0 \therefore 6I_x = 0$$



$$I_N = \frac{20}{4} = 5$$

$$I_N = 5A$$

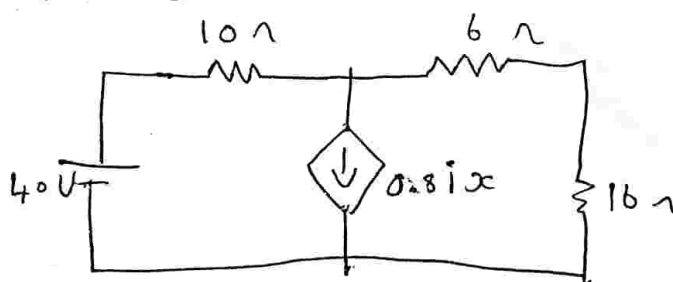
$$R_{TH} = \frac{30}{5}$$

$$R_{TH} = 6\Omega$$

$$I_L = \frac{30}{6+9}$$

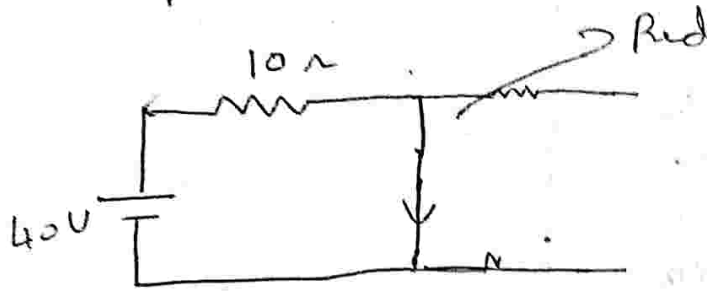
$$I_L = 2A$$

2) Determine the current through 16Ω Resistor using Thevenin theorem



To find V_{TH}
Remove 16Ω

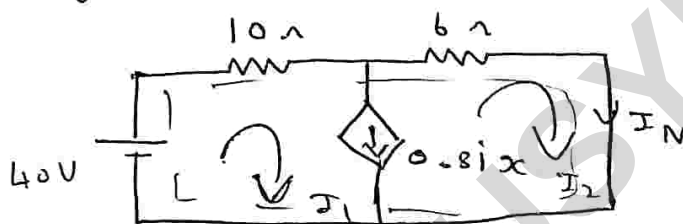
$$\sin \alpha = 0 \quad 0.8 \sin \alpha$$



$$V_{TH} = \frac{40 \times 16}{16}$$

$$V_{TH} = 40V$$

To find



$$40 - 10I_1 - 0.8I_1 + 0.8I_2 = 0$$

$$i_x = I_2$$

$$I_1 - I_2 = 0.8i_x$$

$$I_1 - 1.8I_2 = 0 \quad \text{--- (1)}$$

$$40 - 10I_1 - 6I_2 = 0$$

$$40 = 10I_1 + 6I_2 = 0 \quad \text{--- (2)}$$

$$I_1 = 3$$

$$I_2 = 1.66$$

$$I_2 = I_N = 1.66A$$

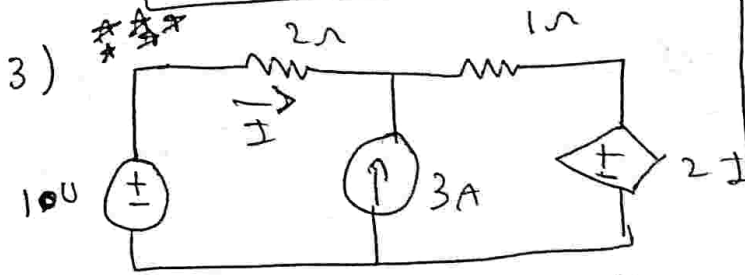
$$R_{TH} = \frac{40}{1.66}$$

$$R_{TH} = 24.09\Omega$$

$$I_L = \frac{V_{TH}}{R_{TH} + R_L}$$

$$I_L = 0.997 \text{ A}$$

Therminis theorem for both dependent and independent sources.

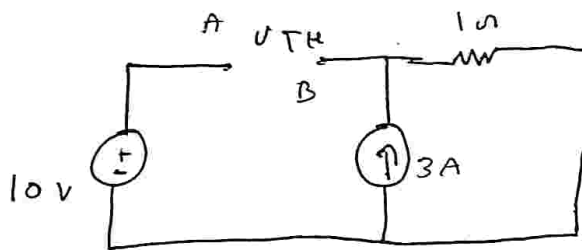


Find the current I for n/w shown using theorem

$$R_L = 2\Omega$$

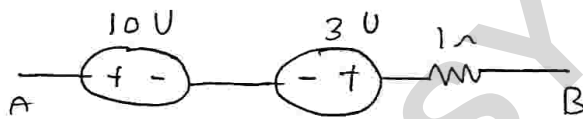
To find V_{th} open load

since 2Ω is removed $I=0$



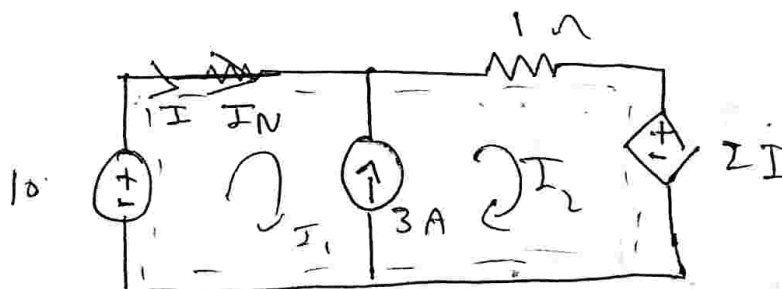
$$\therefore 2I = 0$$

here it is shorted



$$\therefore V_{th} = 7 \text{ V}$$

To find I_N short load mark I_N



$$I = I_1$$

$$I_2 - I_1 = 3 \dots \textcircled{1}$$

appl KVL to super mesh

$$-1I_2 - 2I_1 = -10$$

$$I_1 = 2.33 \text{ A}$$

$$I_2 = 5.33 \text{ A}$$

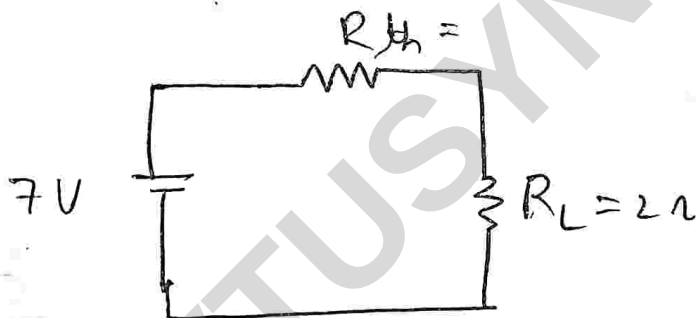
$$\therefore I_N = 2.33 \text{ A}$$

$$R_{th} = \frac{V_{Th}}{I_N} = \frac{7}{2.33}$$

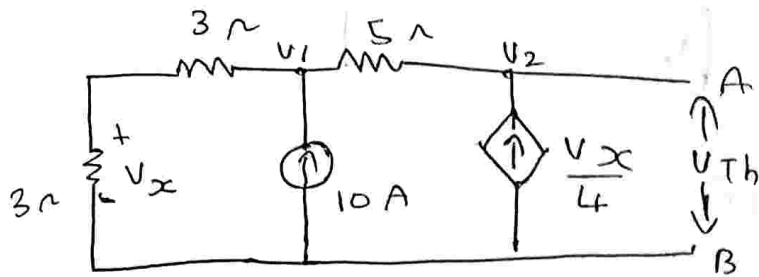
$$R_{th} = 3 \Omega$$

$$I_L = \frac{V_{Th}}{R_{Th} + R_L} = \frac{7}{3+2}$$

$$I_L = 1.4 \text{ A}$$



4) Find the thevenin's equivalent for the circuit shown



apply KCL at node.

$$V_{Th} = V_2$$

$$\frac{V_1}{6} + \frac{V_1 - V_2}{5} = 10$$

$$V_x = 0.5 V_1$$

$$V_1 \left(\frac{1}{6} + \frac{1}{5} \right) - \frac{V_2}{5} = 10$$

$$0.366 V_1 - 0.2 V_2 = 10 \quad \text{--- (1)}$$

$$\frac{V_2 - \frac{V_1}{4}}{5} = \frac{V_x}{4} = 0$$

$$0.2 V_2 - 0.25 V_1 = \frac{0.5 V_1}{4}$$

$$-0.2 V_1 - 0.125 V_1 + 0.2 V_2 = 0 \quad \text{--- (2)}$$

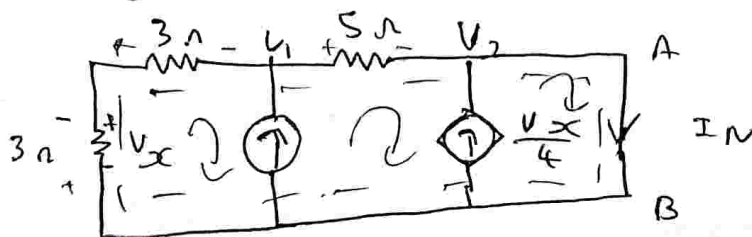
$$-0.325 + 0.2 V_2 = 0 \quad \text{--- (3)}$$

$$V_1 = 240.38$$

$$V_2 = 390.625$$

$$V_{Th} = V_2 = 390.62 V$$

To find I_N short terminals A & B



$$V_x = -I_1 R$$

$$V_x = -I_1 \cdot 3$$

$$V_x = -3I_1$$

app ① Super mesh

$$I_2 - I_1 = 10 \quad \dots \quad (1)$$

② Super mesh.

$$I_3 - I_2 = \frac{V_x}{4}$$

$$I_3 - I_2 = -\frac{3I_1}{4}$$

$$+0.75I_1 - I_2 + I_3 = 0 \quad \dots \quad (2)$$

apply KVL to super mesh.

$$-6I_1 - 5I_2 = 0 \quad \dots \quad (3)$$

$$I_1 = -4.54 \text{ A}$$

$$I_2 = 5.45 \text{ A}$$

$$I_3 = 8.86 \text{ A}$$

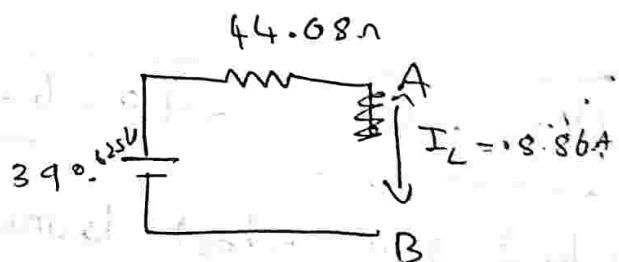
$$\therefore I_N = 8.86 \text{ A}$$

$$R_{th} = \frac{V_{th}}{I_N} = \frac{390.625}{8.86}$$

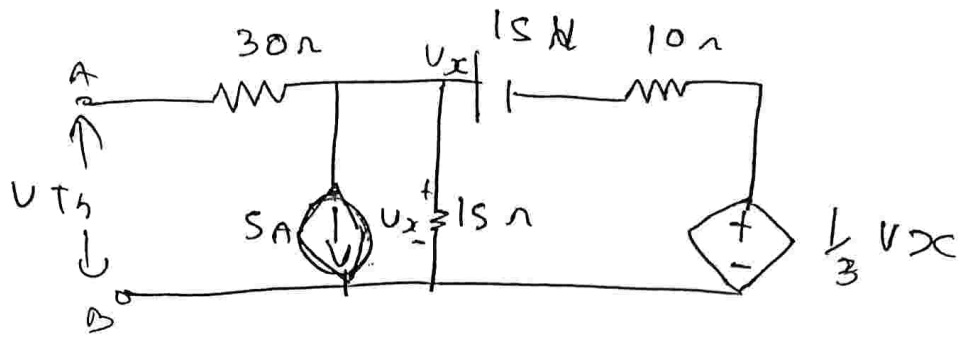
$$R_{th} = 44.08 \Omega$$

$$I_L = \frac{V_{th}}{R_{th}}$$

$$I_L = 8.86 \text{ A}$$

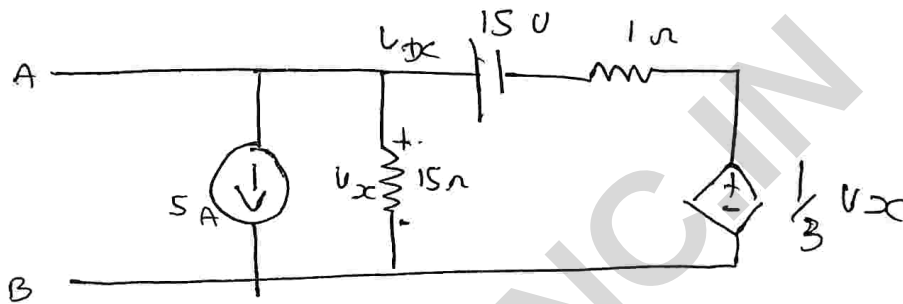


5) obtain the Thevenin equivalent for the ckt shown



To find V_{Th} .

Since the load terminals is open current $I = 0$ hence 30Ω is redundant



apply KCL

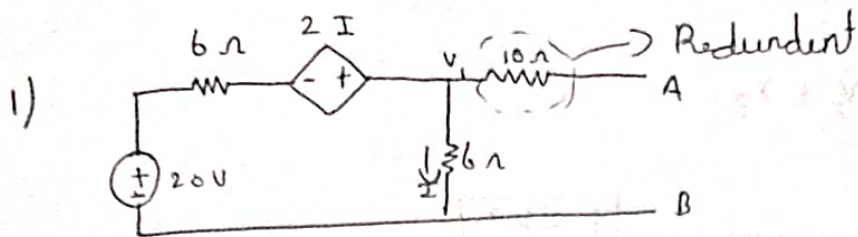
$$\frac{V_x}{15} + \frac{V_x - 15}{10} - \frac{1}{3} V_x = 0 - 5$$

$$V_x \left(\frac{1}{15} + \frac{1}{10} - \frac{1}{3} \right) = 1.5 - 5$$

$$V_x (0.2) = 1.5 - 5$$

$$V_x = \frac{1.5 - 5}{0.2}$$

$$\boxed{V_x = -17.5V}$$



$$V_{Th} = V_1$$

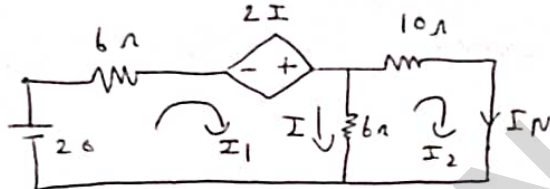
$$20 - 6I_1 + 2I_1 - 6I = 0$$

$$20 - 10I = 0$$

$$I = \frac{20}{10} \text{ A}$$

$$I = 2 \text{ A}$$

$$V_{Th} = I \times R = 2 \times 6 = 12 \text{ V}$$



apply KVL to loop 1

$$I = I_1 - I_2$$

$$-10I_2 - 6(I_2 - I_1) = 0$$

$$-10I_2 - 6I_2 + 6I_1 = 0$$

$$6I_1 - 16I_2 = 0$$

apply KVL to loop 2

$$20 - 6I_1 + 2I - 6(I_1 - I_2) = 0$$

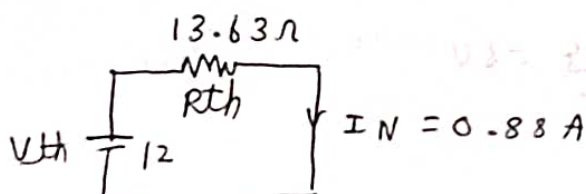
$$20 - 10I_1 + 4I_2 = 0$$

$$10I_1 - 4I_2 = 20$$

$$I_1 = 2.35 \text{ A} \quad I_2 = 0.88 \text{ A}$$

$$I_N = I_2 = 0.88 \text{ A}$$

$$R_{Th} = \frac{12}{0.88} = 13.63 \Omega$$



$$2) V_1 = 20, V_2 = 40, V_3 = 50, Z_1 = 2, Z_2 = 4, Z_3 = 5$$

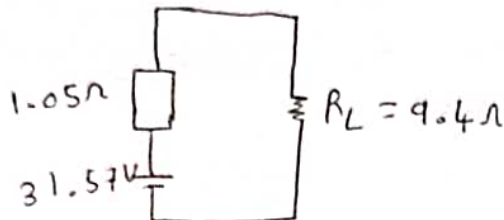
$$Y_1 = 0.5, Y_2 = 0.25, Y_3 = \frac{1}{5} = 0.2$$

$$Z_m = \frac{1}{Y_1 + Y_2 + Y_3}$$

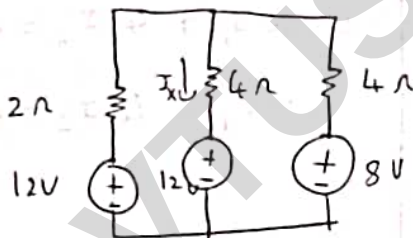
$$Z_m = \frac{1}{0.5 + 0.25 + 0.2} = \boxed{1.05 \Omega}$$

$$V_m = \frac{20(0.5) + 40(0.25) + 50(0.2)}{0.5 + 0.25 + 0.2}$$

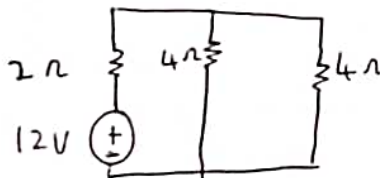
$$\boxed{V_m = 31.57 V}$$



3) Find I_x using Super position theorem:



i) consider 12V alone

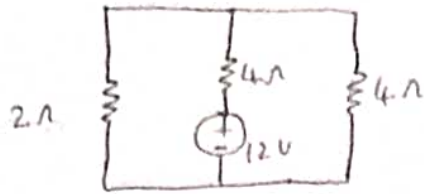


$$4 \parallel 4$$

$$V_1 = \frac{2(12)}{2+2} = \frac{24}{4} = 6V$$

$$I_x^1 = \frac{V}{R} = \frac{6}{4} = 1.5A$$

ii) consider 12V alone

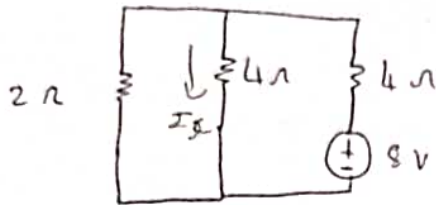


$$4 \parallel 2 = 1.33 \Omega$$

$$I = \frac{V}{R} = \frac{-12}{4 + 1.33}$$

$$I = -2.25 \text{ A}$$

iii) consider 8V alone



$$V = \frac{8 \times 4}{4 + 2}$$

$$4 \parallel 2 = 1.33 \Omega$$

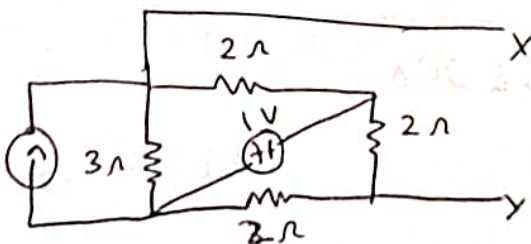
$$V \parallel R = \frac{8(1.33)}{5.33} = 2 \text{ V}$$

$$I = \frac{V}{R} = 0.5 \text{ A}$$

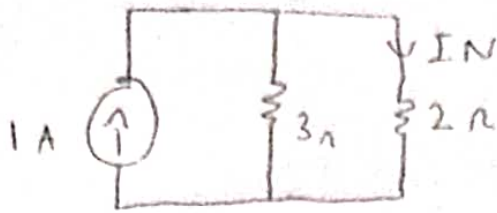
$$I_T = 1.5 - 2.25 + 0.5$$

$$I_T = -0.25 \text{ A}$$

+) Find the current through $R_L = 1 \Omega$

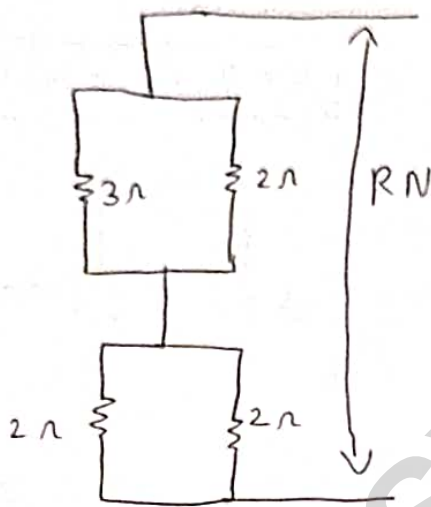


To find I_N , short R_L & replace with I_N



$$I_N = \frac{3(1)}{2+3} = 0.6A$$

To find R_N , open load

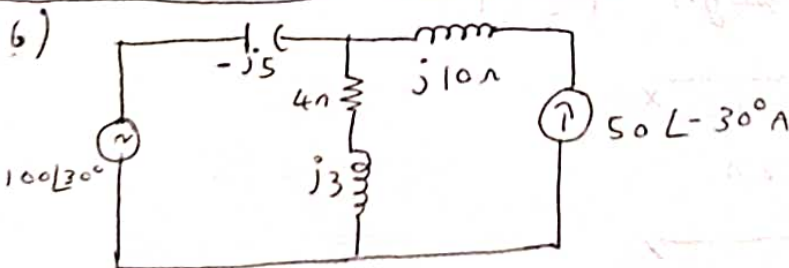


$$R_N = (3 \parallel 2) + (2 \parallel 2) = 2.2 \Omega$$

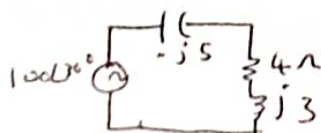
for $R_L = 1 \Omega$

$$I_L = \frac{I_N R_N}{R_N + R_L} = \frac{(0.6 \times 2.2)}{2.2 + 1} = 0.41A$$

$$I_L = 0.41A$$

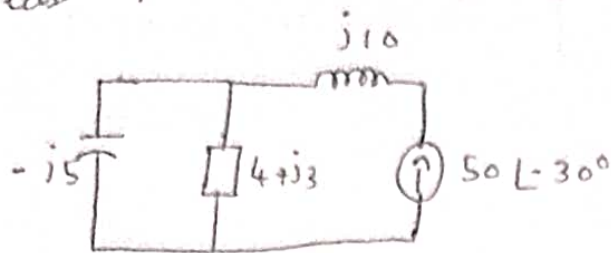


consider only $100 \angle 30^\circ$ alone



$$I' = \frac{V}{Z_T} = \frac{15.42 \cdot 98.8i}{(-j5) + (4+j3)} \Rightarrow 12.96 - 18.2iA$$

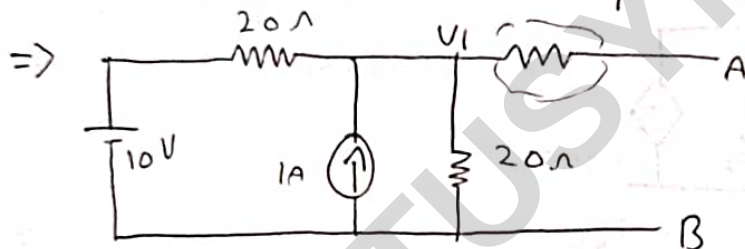
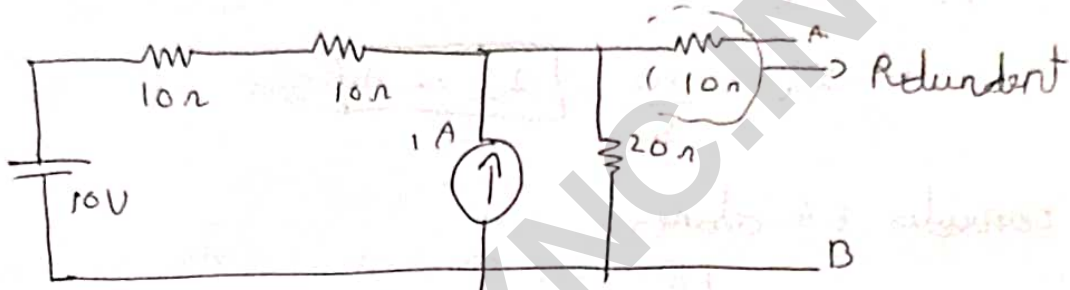
case ii)



$$I(4+j3) = \frac{-j5(50\angle-30^\circ)}{(-j5) + (4+j3)} = 53.25 + 16.9i$$

$$I_T = I' + I'' = 66.21 - 1.3i$$

7)



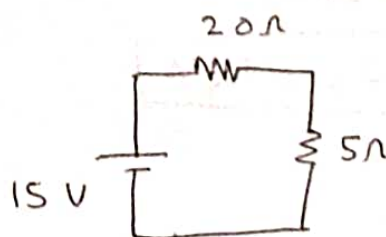
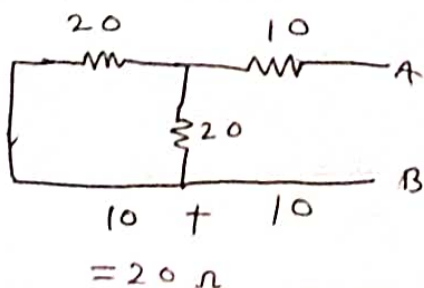
To find V_1 apply KCL

$$\frac{V_1 - 10}{20} - 1 + \frac{V_1}{20} = 0$$

$$0.1V_1 - 1.5 = 0$$

$$V_1 = \frac{1.5}{0.1} \quad \boxed{V_1 = 15V}$$

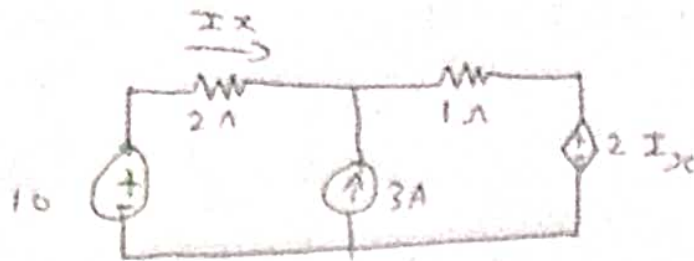
To find R_{th}



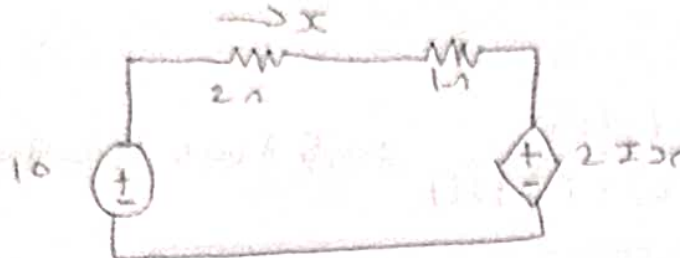
$$V_{across 5\Omega} = \frac{5 \times 15}{25} = 3V$$

$$P_{5\Omega} = \frac{3^2}{5} = 1.8W$$

8) find I_x using superposition theorem



consider 10V alone



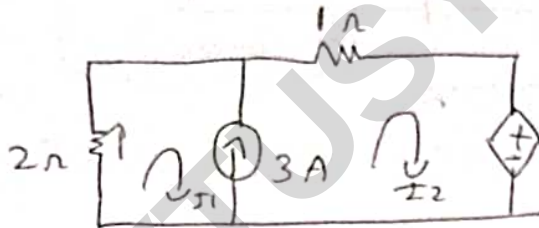
apply KVC

$$10 - 2I_x - 1I_x - 2I_x = 0$$

$$I_x = \frac{10}{5}$$

$$I_x = 2A$$

consider 3A alone



$$I_2 - I_1 = 3 \quad \text{--- (1)}$$

$$-2I_x - 1I_2 - 2I_x = 0$$

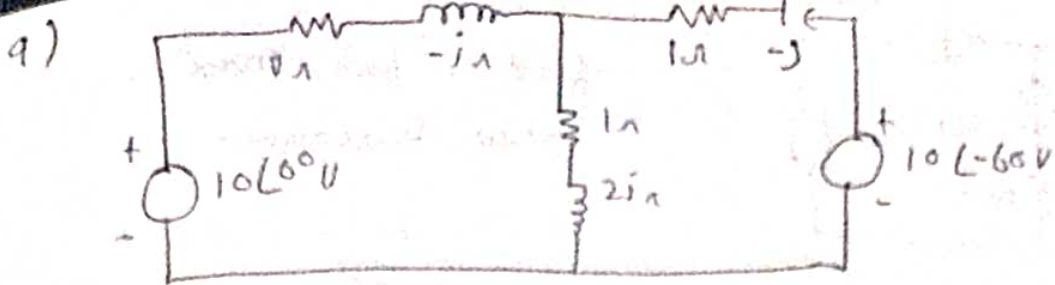
$$-4I_1 - 1I_2 = 0 \quad \text{--- (2)}$$

Solving (1) & (2)

$$I_1'' = -0.6A$$

$$I_{xc} = 2 + (-0.6)$$

$$I_x = 1.4A$$



i) consider 10V

$$R_T = (1 + 2j) \parallel ((1 - j) + (1 - j))$$

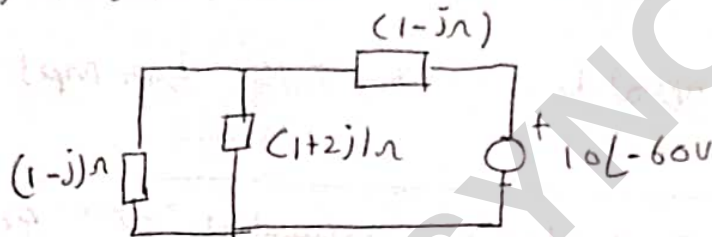
$$R_T = (2.4 - 1.2j) \Omega$$

$$I_T = \frac{10}{2.4 - 1.2j}$$

$$I_{Z_3} = \frac{I_T \cdot Z_2}{Z_2 + Z_3}$$

$$I_T = 3.33 + 1.66j$$

ii) consider $10\angle-60^\circ \Rightarrow -9.52 + 3.04j$



$$R_T = 2.4 - 1.2j \Omega$$

$$I_T = \frac{V}{R} = \frac{-9.52 + 3.04j}{2.4 - 1.2j}$$

$$I_T = -3.68 - 0.573j$$

$$I_T = (3.72 \angle -2.98^\circ) A$$

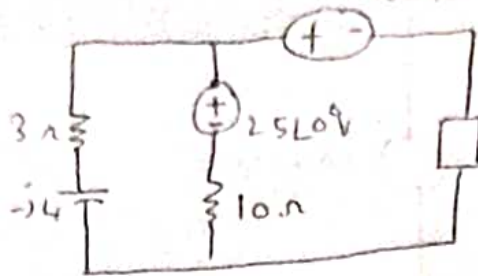
$$I_{Z_3} = \frac{(1-j)(3.72 \angle -2.98^\circ)}{(1-j)(1+2j)}$$

$$I_{Z_3} = 2.35 \angle 2.04^\circ$$

$$I = I_{Z_3}^I + I_{Z_3}^{II}$$

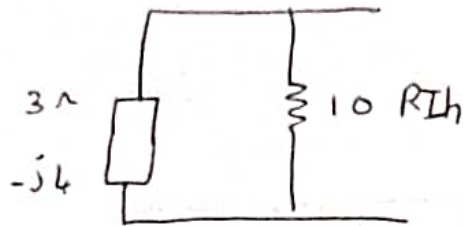
$$I = 0.73 \angle 0.62^\circ A$$

10)



find Z_L for max power transmission.

Z_L should be equal to R_{th} to find R_{th} , open Z_L , short V source

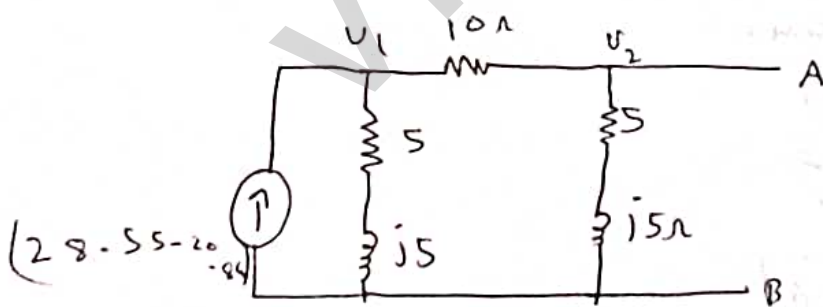


$$R_{th} = (3 - j4) \parallel 10$$

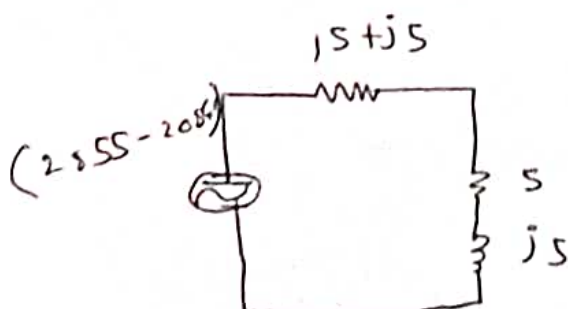
$$R_{th} = 2.97 - 2.16j \Omega$$

R_{th} should be equal to $2.97 - 2.16j$ for mpt

11) obtain Thevenin's & norton's equivalent circuit find the current through 10Ω RL across A-B



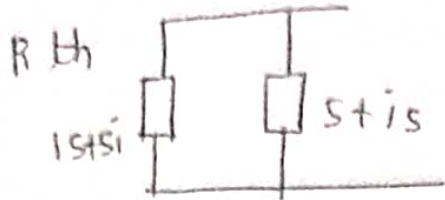
convert to voltage source



$$V_{th} = \frac{(28.55 - 20.841)(5 + j5)}{(5 + j5) + (5 + j5)}$$

$$V_{th} = 11.17 \angle -0.30$$

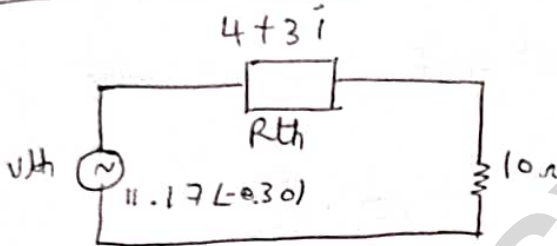
$$V_{th} = 10.64 - 3.39j$$



$$R_{th} = (5 + j5) \parallel (5 + j5)$$

$$R_{th} = (4 + 3j) \Omega$$

$$R_{th} = 5 \angle 0.64 \text{ rad}$$



→ Norton's equivalent circuit.

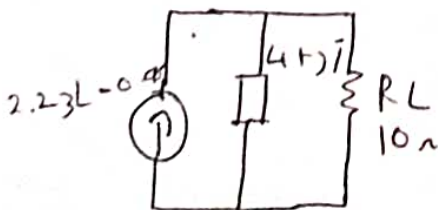
$$R_{th} = \frac{V_{th}}{I_N}$$

$$I_N = \frac{V_{th}}{R_{th}}$$

$$I_N = \frac{10.64 - 3.39j}{5 \angle 0.64}$$

$$I_N = 2.120 - 0.70j$$

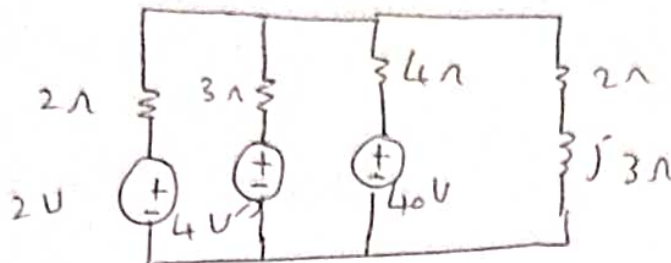
$$R_N = R_{th}$$



$$I_L = \frac{I_N \cdot R_N}{R_N + R_L} = \frac{(1.29 - 18j) (4 + 3j)}{(4 + 3j) (10)}$$

$$I_L = 0.675 - 0.38j \text{ A}$$

12) millman's theorem



$$Z_1 = 2\Omega \quad Z_2 = 3\Omega \quad Z_3 = 4\Omega$$

$$y_1 = \frac{1}{Z_1} = 0.5 \text{ S} \quad y_2 = \frac{1}{Z_2} = 0.33 \text{ S} \quad y_3 = \frac{1}{Z_3} = 0.25 \text{ S}$$

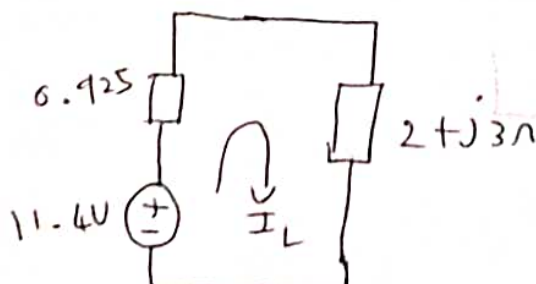
$$Z_m = \frac{1}{0.5 + 0.33 + 0.25}$$

$$Z_m = 0.925\Omega$$

$$V_m = \frac{V_1 y_1 + V_2 y_2 + V_3 y_3}{y_1 + y_2 + y_3}$$

$$V_m = \frac{(2)(0.5) + 4(0.33) + (40)(0.25)}{0.5 + 0.33 + 0.25}$$

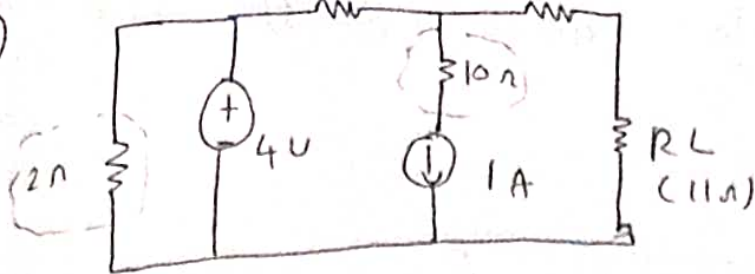
$$V_m = 11.4 \text{ V}$$



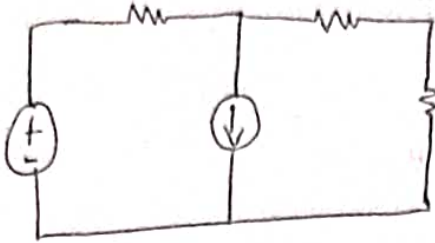
$$I_L = \frac{V_m}{Z_m + 2 + j3} = \frac{11.4}{0.925 + 2 + j3}$$

$$I_L = 1.899 - 1.948j \text{ A}$$

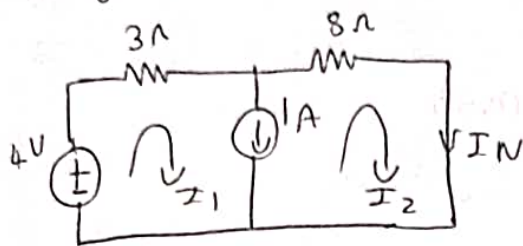
13)



2Ω & 10Ω are redundant from the ckt



To find I_N



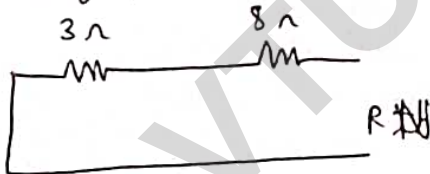
from the ckt $I_2 = I_N$

from ① & ②

$$I_1 = 1.09A$$

$$I_2 = 0.09A$$

To find R_N



$$R_{th} = 3 + 8$$

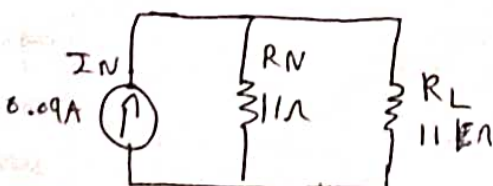
$$R_{th} = 11\Omega$$

$$I_L = \frac{I_N \cdot R_N}{R_N + R_L}$$

$$I_L = \frac{(0.09)(11)}{11 + 11}$$

$$11 + 11$$

$$I_L = 0.045A$$



supermesh eqn.

$$1 = I_1 - I_2 \dots \textcircled{1}$$

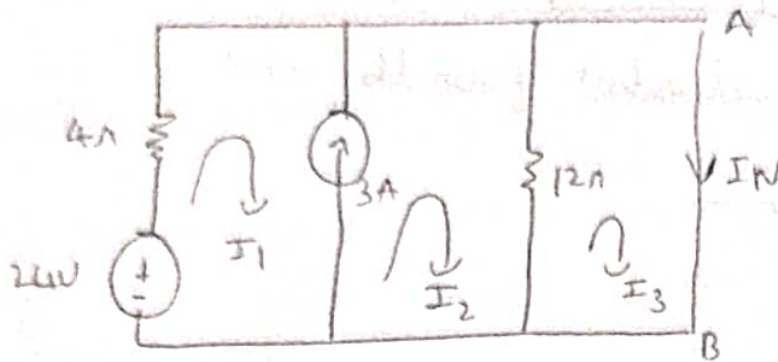
apply KVL to supermesh

$$4 - 3I_1 - 8I_2 = 0$$

$$3I_1 + 8I_2 = 4 \dots \textcircled{2}$$

23) Find the Norton's equivalent for the circuit shown.

12)



from the circuit $I_3 = I_N$

$$I_2 - I_1 = 3 \dots \textcircled{1}$$

applying KVL to super mesh

$$24 - 4I_1 - 12I_2 + 12I_3 = 0$$

$$-4I_1 - 12I_2 + 12I_3 = -24 \dots \textcircled{2}$$

applying KVL to loop ③

$$-12I_3 + 12I_2 = 0$$

$$12I_2 - 12I_3 = 0 \dots \textcircled{3}$$

solving ①, ②, ③

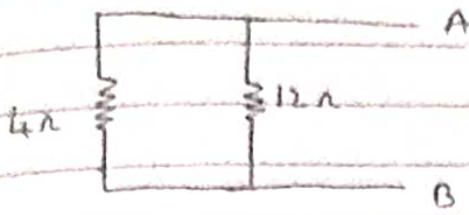
$$I_1 = 6A$$

$$I_2 = 9A$$

$$I_3 = 9A$$

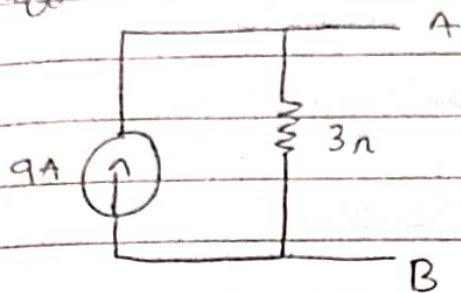
$$\therefore I_3 = I_N = 9A$$

To find R_N

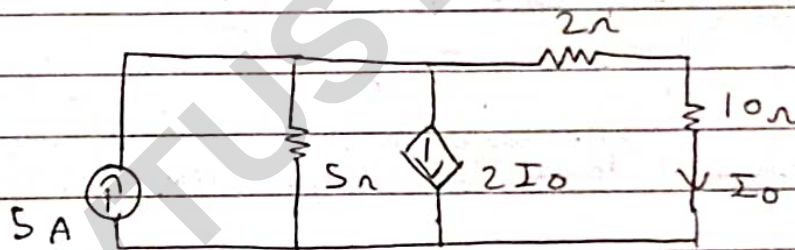


$$R_N = 3 \Omega$$

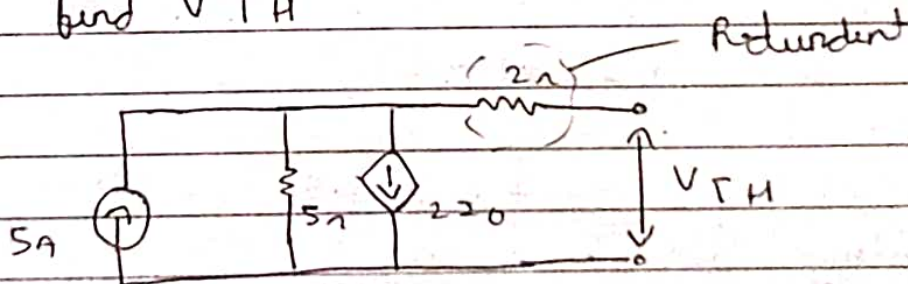
equivalent circuit of Norton's



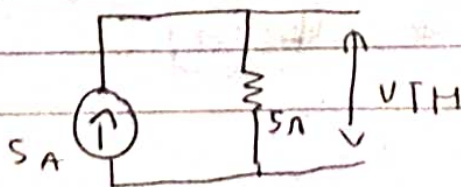
25. In the circuit shown, find the current through 10Ω resistor using Thevenin's theorem



→ To find V_{TH}



here R_L is open so $I_0 = 0$ hence $2I_0 = 0$

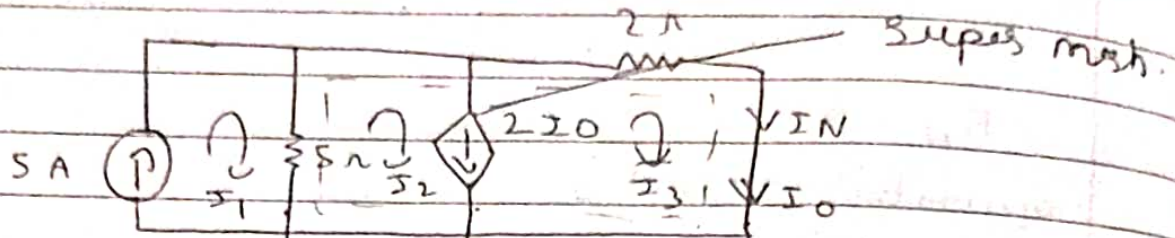


from ohm's law

$$V = IR = \frac{5 \times 5}{V = 25V}$$

$$V_{TH} = 25V$$

To find I_N



Super mesh equation

$$I_2 - I_3 = 2I_3$$

$$\therefore I_3 =$$

$$I_2 - I_3 = 2I_3$$

$$I_2 - 3I_3 = 0 \quad \dots (1)$$

from the ckt

$$I_1 = 5A$$

applying KVL to super mesh.

$$-5I_2 + 5I_1 - 2I_3 = 0$$

$$-5I_2 + 5(5) - 2I_3 = 0$$

$$-5I_2 - 2I_3 = -25 \quad \dots (2)$$

solving (1) & (2)

$$I_2 = 4.473A$$

$$I_3 = 1.473A$$

$$\therefore I_N = I_3 = 1.473A$$

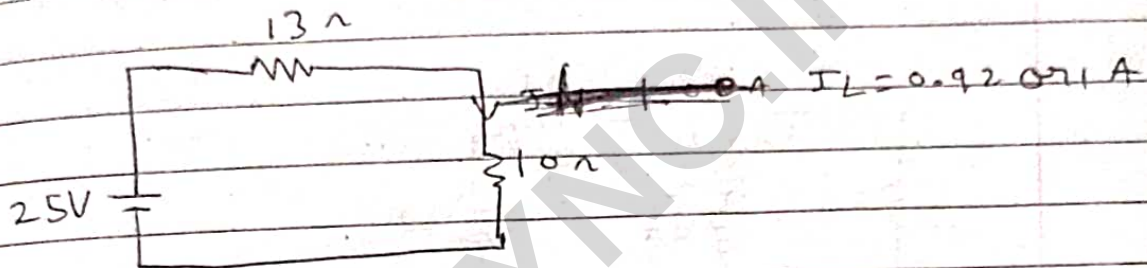
$$I_N = 1.473A$$

$$R_{TH} = \frac{25}{1.473}$$

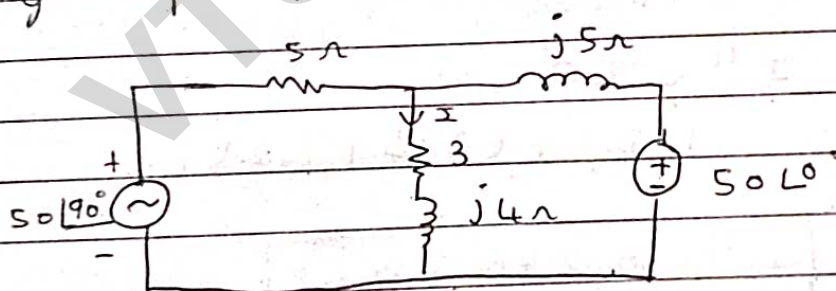
$$R_{TH} = 17 \Omega$$

$$I_L = \frac{25}{17 + 10}$$

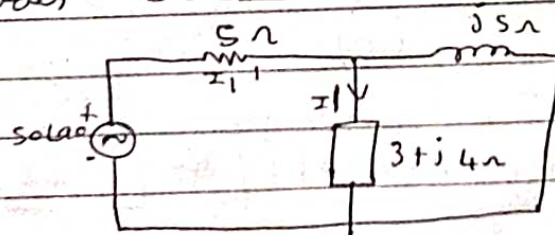
$$I_L = 0.92 \approx 1 \text{ A}$$



26. Find the current through $(3 + j4) \Omega$ using super position theorem



Consider $50 \angle 90^\circ$



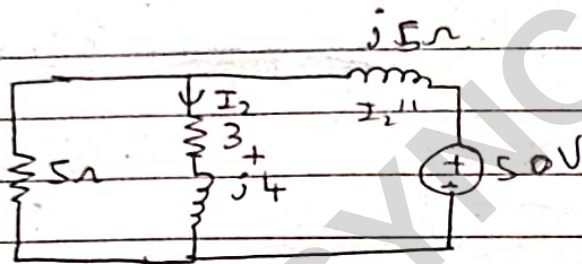
$$I_1 = \frac{I_1' \times j5}{(3 + j4) + (j5)}$$

$$I_1' = \frac{5 \angle 90^\circ}{5 + (0.833 + 2.5i)}$$

$$I_1' = 3.10 + 7.241i$$

$$I_1 = \frac{(3.10 + 7.241i) \times (j5\Omega)}{(3 + j4) + (j5)}$$

$$I_1 = 1.32 + 0.098i \quad \text{or} \quad I_1 = 1.33 \angle 4.22^\circ$$



$$I_2 = \frac{5 \times I_2''}{(3 + j4) + (j5)}$$

$$I_2'' = \frac{j50}{(j5\Omega) + (2.5 + 1.25i)}$$

$$I_2'' = 0.275 - 0.68i$$

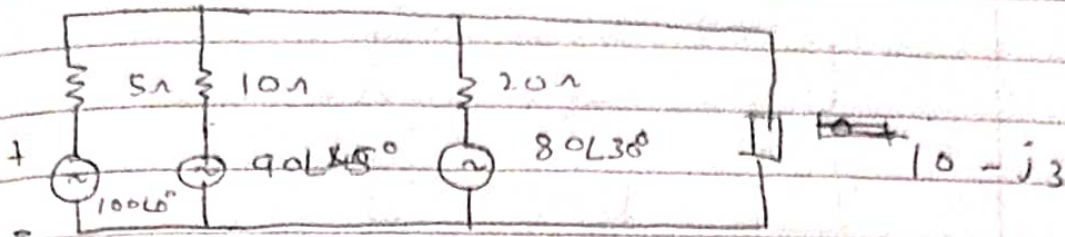
$$I_2 = \frac{5 \times (0.275 - 0.68i)}{(3 + j4) + (j5)}$$

$$I_2 = -0.202 + 0.101i \quad \text{or} \quad 0.22 \angle 153.43^\circ \text{ A}$$

$$I = I_1 + I_2$$

$$\therefore I_1 = 1.14 \angle 9.85^\circ \text{ A} \quad \text{or} \quad 1.12 + 0.196i \text{ A}$$

27) for the ckt shown find the current flowing in $10-j3$ using millman's theorem



$$Z_1 = 5$$

$$Z_2 = 10$$

$$Z_3 = 20$$

$$Y_1 = 0.2$$

$$Y_2 = 0.1$$

$$Y_3 = 0.05$$

$$Z_m = \frac{1}{Y_1 + Y_2 + Y_3} = \frac{1}{0.2 + 0.1 + 0.05} = 2.85$$

$$Z_m = 2.85 \Omega$$

$$V_1 = 100$$

$$V_2 = 63.63 + 63.63j$$

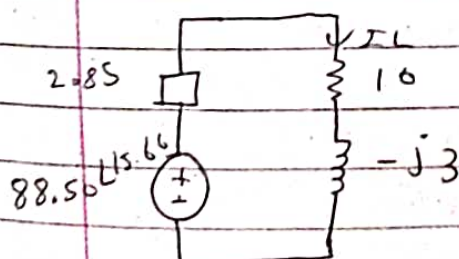
$$V_3 = 69.28 + 40j$$

$$V_m = \frac{V_1 Y_1 + V_2 Y_2 + V_3 Y_3}{Y_1 + Y_2 + Y_3}$$

$$V_m = \frac{100(0.2) + 63.63 + 63.63j(0.1) + 69.28 + 40j(0.05)}{0.2 + 0.1 + 0.05}$$

$$V_m = 85.22 + 23.89j$$

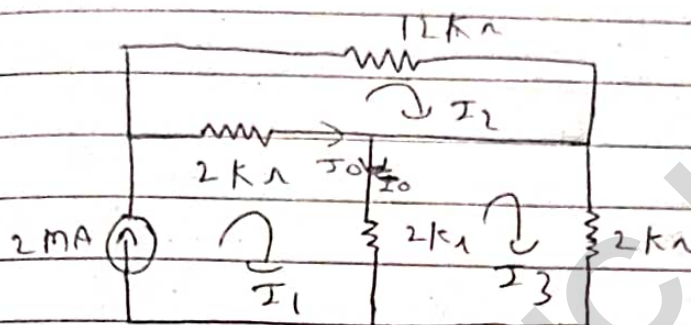
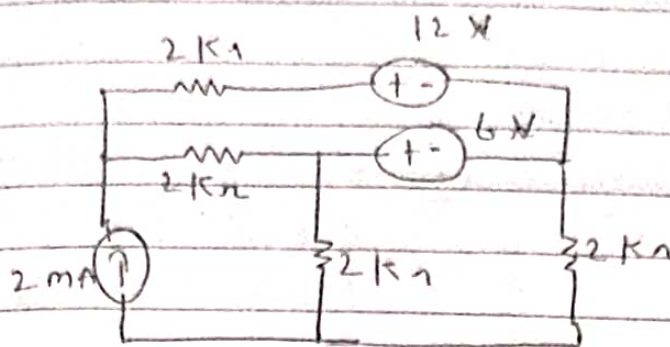
$$\text{or } V_m = 88.50 \angle 15.66^\circ$$



$$I_L = 6.70 \angle 28.80^\circ \text{ A}$$

$$\text{or } I_L = 5.877 + 3.23j \text{ A}$$

29. using superposition theorem to find i_o in the ckt



$$-4I_1 + 2I_3 = 2 \quad \dots \textcircled{1}$$

$$-12I_2 - 12I_2 + 12I_1 = 0 \quad \dots \textcircled{2}$$

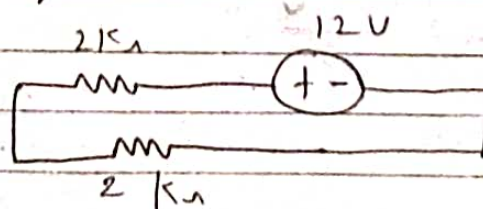
$$-2I_3 - 2I_1 - 2I_3 = 0 \quad \dots \textcircled{3}$$

$$I_1 = -0.4 \text{ mA} \quad I_2 = -0.2 \text{ mA} \quad I_3 = 0.2 \text{ mA}$$

$$I_1 = I_1' = -0.4 \text{ mA}$$

Case ②

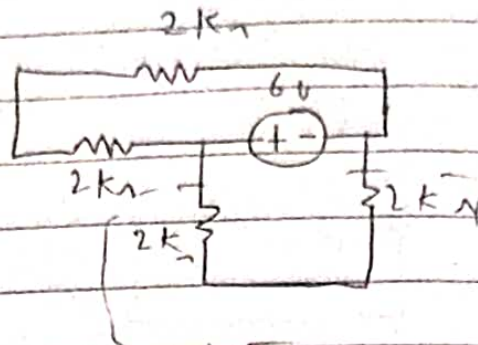
12V acting alone.



$$I = \frac{V}{R} = \frac{12}{4 \text{ k}}$$

$$I_1'' = 3 \text{ mA}$$

Working alone:



Reduced
as they are
in series with

$$I = \frac{V}{R} = \frac{6}{4} = 1.5 \text{ mA}$$

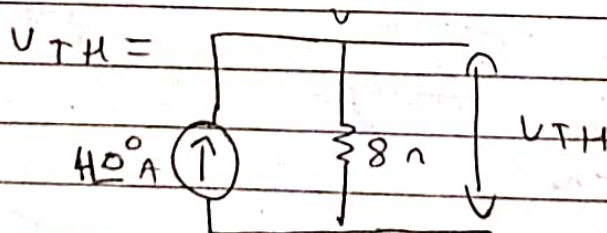
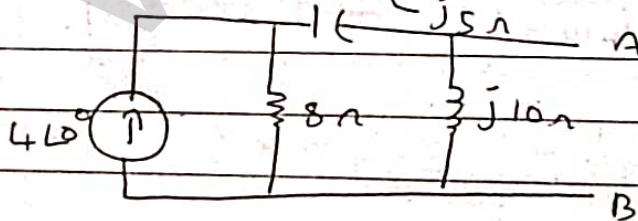
$$I_1''' = 1.5 \text{ mA}$$

$$I_0 = I_1' + I_1'' + I_1'''$$

$$I_0 = -0.4 \text{ mA} + 3 \text{ mA} + 1.5 \text{ mA}$$

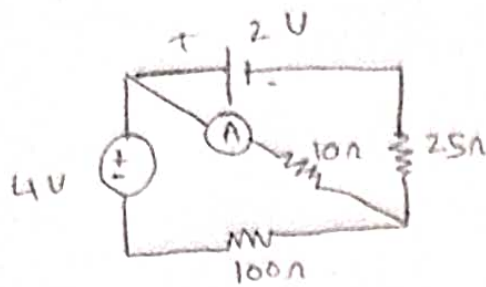
$$I_0 = 4.1 \text{ mA}$$

30) Find the thevenin & norton's equivalent at a

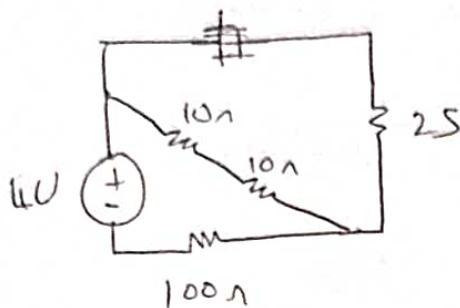


$$V_{TH} = 32 \text{ V}$$

21) Determine I through an Ammeter having internal resistance



→ consider current through ammeter as I
 case 1: 4V acting alone.



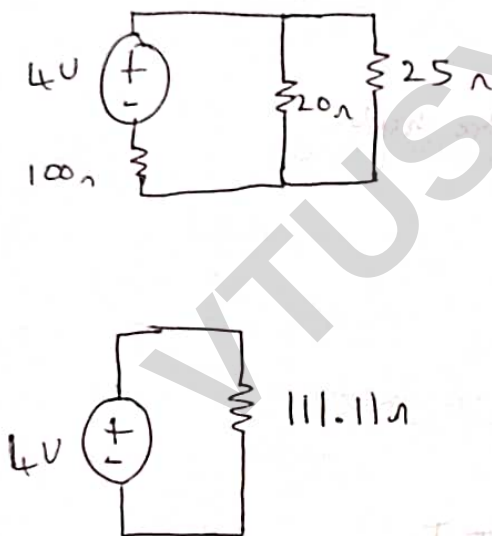
$$R = 10 + 10 \\ = 20\Omega$$

$$R = 20 || 25$$

$$R = 11.11\Omega$$

$$R = 11.11 + 100$$

$$R = 111.11$$



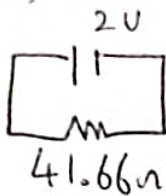
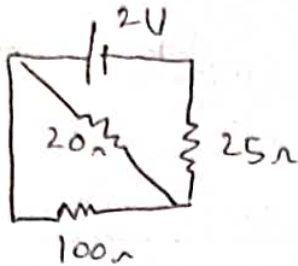
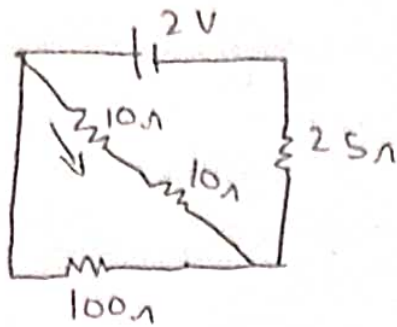
$$I = \frac{V}{R} = \frac{4}{111.11} = 0.036 A$$

Apply current divider rule.

$$I_1 = \frac{I \times 25}{20 + 25}$$

$$I_1 = 20 mA$$

case 2: 2V acting alone.



$$I = \frac{2}{41.66}$$

$$I = 48 \text{ mA}$$

applying current divider rule.

$$I_2 = \frac{I \times 100}{20 + 100}$$

$$I_2 = 40 \text{ mA}$$

$$\begin{aligned} \text{Current through} &= I_1 + I_2 \\ \text{ammeter} &= 20 \text{ mA} + 40 \text{ mA} \end{aligned}$$

$$I = 60 \text{ mA}$$